

Chair of Renewable and Sustainable Energy Systems

Technische Universität München



**Building Energy Management System
Based on Multi-Agent Deep
Reinforcement Learning**

Master's Thesis submitted for the degree of Master of Science
(M.Sc.) in Energy and Process Engineering

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Contents

1	System Modelling and Problem Formulation	1
1.1	Overall System	1
1.2	Load of the Prosumers	4
1.3	PV Modelling	4
1.4	Stationary Battery Energy Storage Model	5
1.5	EV Modelling	6
1.5.1	EV Connection Window	6
1.5.2	EV arrival SoC and Charging Power Constraints	7
1.5.3	EV SoC Status	8
1.5.4	EV Departure SoC Requirement	8
1.6	Electricity Price Model	9
1.7	Net Load, Power Flow, and Grid Safety	9
1.7.1	Voltage Constraints	10
1.7.2	Line Loading Constraints	10
1.7.3	Transformer Loading Constraints	11
1.8	Optimization Objective	11
2	Methodology	13
2.1	Overall Framework	13
2.2	LSTM Forecasting	14
2.3	MADRL Observation	15
2.3.1	Local Observation	16
2.3.2	Forecast Sequence Observation	16
2.3.3	Safety Observation	16
2.4	MADRL Action Design	17
2.4.1	Battery Action Mapping	17
2.4.2	PV Action Mapping	18
2.4.3	EV Action Mapping	18
2.5	Reward Function Design	18
2.5.1	EV Charging Cost	19
2.5.2	EV Costweight Modelling	19
2.5.3	EV Departure Penalty	19

2.5.4	EV Progress Penalty	20
2.5.5	Price-Aware Penalty	20
2.5.6	Battery Reward and Regularization	21
2.5.7	Grid Safety Penalty	22
2.5.8	Total Reward	22
2.6	Safety Projection for Grid Constraints	22
2.7	EV Departure Requirement Handling	24
2.7.1	Soft Mode	24
2.7.2	Hard Mode	24
2.7.3	Emergency Charging	25
2.8	MADRL Training Algorithm	25
2.8.1	Training Pseudocode	26
2.9	Baseline Controllers	27
2.9.1	Local MPC	27
2.9.2	ADMM-MPC	28
2.9.3	Rule-Based Controllers	29
3	Experimental Setup	31
3.1	Dataset and Simulation Period	31
3.2	Battery and EV Parameters	32
3.3	Training Parameters	33
3.3.1	LSTM Training Parameters	33
3.3.2	MADRL Training Parameters	33
3.4	Network Configuration	34
3.5	Evaluation Metrics	34
3.5.1	Economic Metrics	34
3.5.2	Grid Safety Metrics	35
3.5.3	EV Departure SoC Metrics	35
3.5.4	Relationship between EV/Battery and Electricity Prices	36
4	Results and Discussion	37
4.1	LSTM Price Prediction Results	37
4.2	MADRL Convergence Analysis	38
4.3	EV soft constraint results	40
4.4	EV hard constraint results	49
4.5	MADRL Results	55

1 System Modelling and Problem Formulation

In this chapter, the multi-prosumer building energy management problem investigated in this thesis is defined, and the physical models and mathematical formulations involved in the environment are presented. The system under study consists of three prosumers connected to a low-voltage distribution network. Each prosumer is equipped with local electrical loads, photovoltaic (PV) generation, a stationary battery energy storage system (BESS), and residential electric vehicle (EV) charging demand. At each discrete time step, the control system determines the battery charging/discharging power, PV utilization, and EV charging power to minimize the total electricity purchasing cost while satisfying both the physical constraints of the devices and the operational safety constraints of the distribution network.

This chapter focuses only on the system modeling and problem formulation, including the system constraints and the optimization objective function. The details of the control methodology, such as how reinforcement learning actions are generated, how the EV control strategies are implemented at the control layer, and how the grid safety projection modifies the control actions, are discussed in Chapter 4.

1.1 Overall System

The system investigated in this thesis addresses the energy management problem of a low-voltage distribution network integrated with distributed energy resources. The system consists of a low-voltage distribution network, 3 prosumer agents, household loads, PV generation, a stationary battery energy storage system, EV charging, electricity price signals, and a power flow calculation module. Each prosumer acts not only as an electricity consumer but also as an electricity producer and the owner of dispatchable energy resources. The primary objective of the system is to coordinate the resources of the three prosumers to minimize the overall operating cost while satisfying both the device operational constraints and the distribution network safety constraints.

The distribution network topology is adopted from SimBench, with the fixed network configuration

1-LV-rural1--0-sw.

The system consists of $N = 3$ prosumers connected to the distribution network buses

$$\mathcal{B} = \{12, 4, 2\}.$$

Let $i \in \mathcal{I} = \{1, 2, 3\}$ denote the prosumer index, and $t \in \mathcal{T} = \{0, \dots, T - 1\}$ represent the discrete time step within an episode. The simulation uses a 15-minute resolution, so a day can be represented as

$$\Delta t = 0.25 \text{ h}, \quad T = 96.$$

One episode corresponds to the operation of a single day. During training, multiple consecutive days can be concatenated to form a longer training window.

The local power profile of each prosumer i consists of external inputs and three kinds of controllable or semi-controllable energy resources. The external inputs include the household load $L_{i,t}^{hh}$, the available PV generation $G_{i,t}$, and the electricity price p_t . In this thesis, only the household load is considered as the base electrical demand. The heat pump load is not included in the current model but can be incorporated as an additional load in future work:

$$L_{i,t} = L_{i,t}^{hh}.$$

The controllable resources include the charging and discharging power of the stationary battery $P_{i,t}^b$, the utilized PV generation $G_{i,t}^{use}$, and the EV charging power $P_{i,t}^{ev}$. Accordingly, the electricity-related variables of each prosumer can be expressed as

$$\{L_{i,t}, G_{i,t}, P_{i,t}^b, P_{i,t}^{ev}\},$$

where $L_{i,t}$ denotes the total local electrical load, $G_{i,t}$ denotes the available PV generation, $P_{i,t}^b$ denotes the stationary battery power, and $P_{i,t}^{ev}$ denotes the EV charging power. In this thesis, the following sign convention is adopted for the battery power: $P_{i,t}^b > 0$ indicates battery charging, whereas $P_{i,t}^b < 0$ indicates battery discharging.

$$-P_i^{b,\max} \leq P_{i,t}^b \leq P_i^{b,\max}.$$

Since only EV charging is considered in this thesis, while vehicle-to-grid is not allowed, the EV charging power satisfies

$$P_{i,t}^{ev} \geq 0.$$

At each time step, the power of all devices within a prosumer is aggregated to determine the net load injected into the distribution network:

$$P_{i,t}^{net} = L_{i,t} - G_{i,t}^{use} + P_{i,t}^b + P_{i,t}^{ev}.$$

When $P_{i,t}^{net} > 0$, the prosumer imports electricity from the distribution grid; when $P_{i,t}^{net} < 0$, the prosumer exports power back to the grid. The net loads of the three prosumers are injected into the low-voltage distribution network model, where power flow analysis is performed using pandapower. The power flow calculation returns the bus voltages, line loading ratios, and transformer loading ratios, which are subsequently used to determine whether the control actions satisfy the distribution network safety constraints.

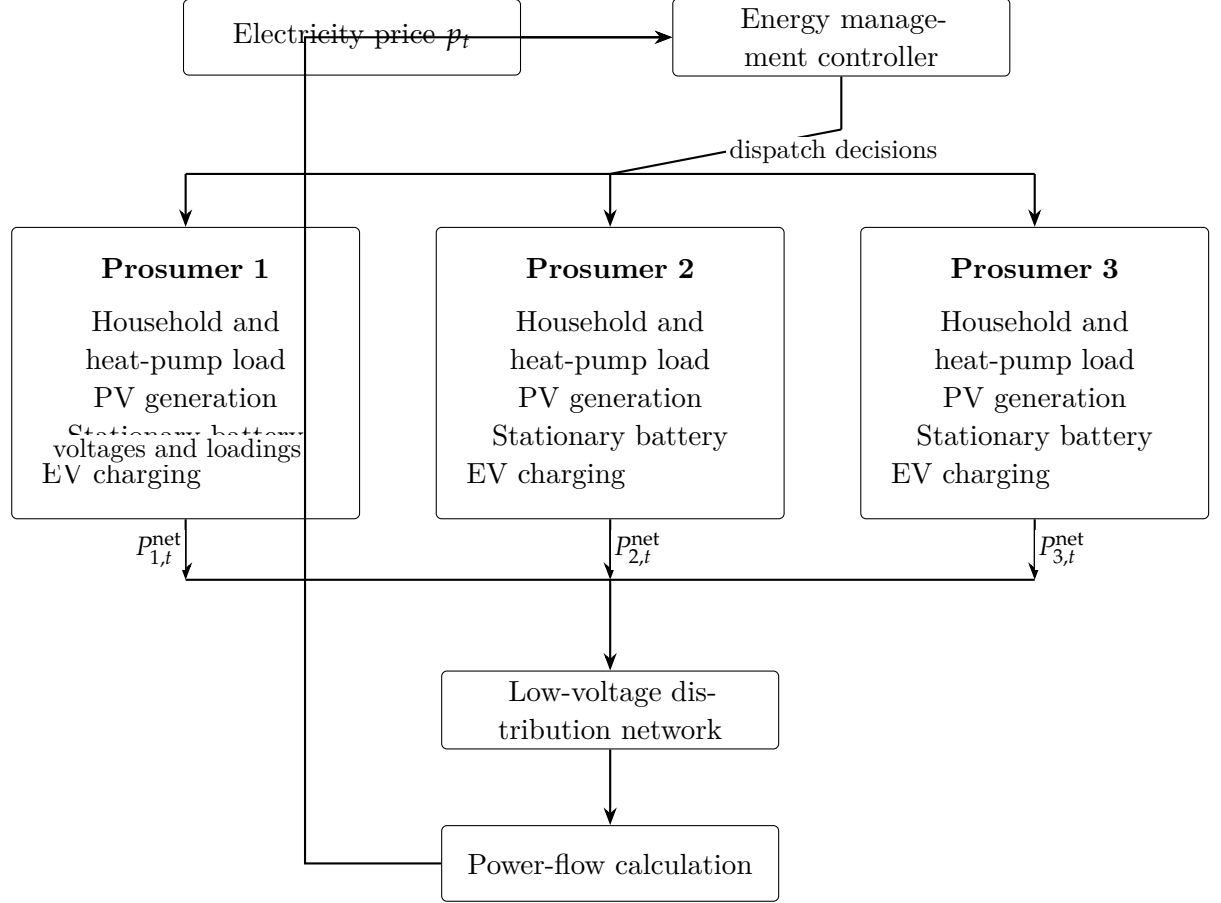


Figure 1.1: Overall architecture of the system investigated in this thesis.

Figure 1.1 illustrates the overall architecture of the system. The household load, heat pump load, PV generation, and electricity price are provided as time-series inputs from the dataset. The stationary battery and EV charging serve as the dispatchable resources on the prosumer side. The grid power flow calculation module maps the net load of each prosumer to the corresponding network operating states, including bus voltages and equipment loading conditions. Accordingly, the problem addressed in this thesis can be

formulated as follows: at each discrete time step, the control system determines $P_{i,t}^b$, $G_{i,t}^{use}$, and $P_{i,t}^{ev}$ for each prosumer based on the load demand, PV generation, electricity price, and the state of charge of the energy storage devices, with the objective of minimizing the total operating cost while ensuring the safe operation of the stationary battery, the EV, and the distribution network.

1.2 Load of the Prosumers

The base electrical load of each prosumer is represented by the household load obtained from the original dataset. The corresponding data file is located at

`datasets/prosumer/household.csv`.

The load profiles of the three prosumers are

SFH12, SFH18, SFH20.

Here, SFH denotes the load profile of a single-family house. The original dataset is read using UTC timestamps and subsequently converted to the Europe/Berlin time zone to ensure that the EV arrival and departure times are consistent with the local time.

Let $L_{i,t}^{hh,raw}$ denote the original household load. In the implementation, an agent-specific scaling factor s_i^L is applied to adjust the original load to the scale used in this study. The default setting is

$$s_1^L = s_2^L = s_3^L = 20.$$

Accordingly, the total electrical load of prosumer i at time step t is given by

$$L_{i,t} = s_i^L \left(L_{i,t}^{hh,raw} \right).$$

This load represents only the external electricity demand of the prosumer and does not include PV generation, battery charging/discharging, or EV charging. These components are incorporated later.

1.3 PV Modelling

The available PV power is obtained from the data file

`datasets/prosumer/pv_reference.csv`.

In the current configuration, the reference column `ref_south` is used. The same reference PV profile is assigned to all three prosumers and scaled by an agent-specific scaling factor s_i^G ,

$$s_1^G = s_2^G = s_3^G = 1.$$

Accordingly, the available PV generation is given by

$$G_{i,t} = s_i^G G_t^{ref}.$$

The controller does not directly modify the available PV generation $G_{i,t}$; instead, it determines the utilized PV power $G_{i,t}^{use}$. Let $u_{i,t}^{pv} \in [0, 1]$ denote the PV utilization factor. Then,

$$\begin{aligned} G_{i,t}^{use} &= u_{i,t}^{pv} G_{i,t}, \\ G_{i,t}^{curt} &= G_{i,t} - G_{i,t}^{use}. \end{aligned}$$

$G_{i,t}^{curt}$ denotes the curtailed PV power. In the current model, no explicit penalty is assigned to PV curtailment. Therefore, PV curtailment primarily affects the system performance indirectly by changing the net load and the operating state of the distribution network.

1.4 Stationary Battery Energy Storage Model

Each prosumer is equipped with a stationary battery. Let C_i^b denote the battery capacity, and let $SOC_{i,t}^b$ denote the battery state of charge (SoC) at the beginning of time step t . The default parameters are

$$C_i^b = 100 \text{ kWh}, \quad SOC_{\min}^b = 0.05, \quad SOC_{\max}^b = 0.95.$$

The battery charging and discharging efficiency is

$$\eta_b = 0.95.$$

Maximum charging and discharging powers are determined by the capacity and maximum charging rate:

$$P_i^{b,\max} = r_{\max}^b C_i^b,$$

where the default $r_{\max}^b = 0.5$, thus a 100 kWh battery corresponds to

$$P_i^{b,\max} = 50 \text{ kW}.$$

The battery power constraints are

$$-P_i^{b,\max} \leq P_{i,t}^b \leq P_i^{b,\max}.$$

The battery SoC dynamics are determined by the charging and discharging directions as mentioned before. If $P_{i,t}^b \geq 0$, the battery is in charging state; if $P_{i,t}^b < 0$, the battery is in discharging state. The SoC is updated as

$$SOC_{i,t+1}^b = \begin{cases} SOC_{i,t}^b + \frac{P_{i,t}^b \Delta t \eta_b}{C_i^b}, & P_{i,t}^b \geq 0, \\ SOC_{i,t}^b + \frac{P_{i,t}^b \Delta t}{\eta_b C_i^b}, & P_{i,t}^b < 0. \end{cases}$$

subject to the boundary constraint

$$SOC_{\min}^b \leq SOC_{i,t+1}^b \leq SOC_{\max}^b.$$

This formulation captures the effect of charging and discharging efficiency losses. During charging, only a fraction η_b of the input electrical energy is stored in the battery. During discharging, a larger amount of internal battery energy must be consumed to deliver the same amount of electrical energy to the external system.

1.5 EV Modelling

Each prosumer is equipped with one residential EV. In the distribution network model, the EV is represented as a controllable charging load characterized by a connection time window, a charging power limit, a battery capacity, and a required departure SoC. In this thesis, vehicle-to-grid (V2G) operation is not considered. Therefore, the EV cannot discharge power to either the building or the distribution network and can only draw electrical energy while connected to the residential charger. Let C_i^{ev} denote the EV battery capacity, and let $SOC_{i,t}^{ev}$ denote the state of charge (SoC) of the EV at the beginning of time step t . The default parameters are

$$C_i^{ev} = 60 \text{ kWh}, \quad SOC_{\min}^{ev} = 0.10, \quad SOC_{\max}^{ev} = 0.95.$$

The maximum power of the residential charger is

$$P_i^{ev,\max} = 11 \text{ kW},$$

and the EV charging efficiency is

$$\eta_{ev} = 0.95.$$

Since the simulation time step is $\Delta t = 0.25 \text{ h}$, the maximum amount of energy that can be charged during a single 15-minute time step is

$$E_{i,\Delta t}^{ev,\max} = P_i^{ev,\max} \Delta t \eta_{ev}.$$

1.5.1 EV Connection Window

EV's connection status is determined by the intraday time steps. The current assumption is that the vehicle arrives home at 18:00 and connects to the residential charger, and departs at 07:00 the next day. Since there are 96 fifteen-minute time steps in a day, the arrival and departure times are

$$t^{arr} = 72, \quad t^{dep} = 28.$$

Let $t_d = t \bmod T$ denote the intraday time step. Since $t^{arr} > t^{dep}$, the connection window spans midnight, and thus a complete parking window consists of the time steps $72, \dots, 95$ from the evening of the current day and the time steps $0, \dots, 27$ from the early morning of the next day:

$$\mathcal{T}^{ev} = \{t : t_d \geq t^{arr}\} \cup \{t : t_d < t^{dep}\}.$$

The connecting status of the EV is represented by the variable A_t^{ev} :

$$A_t^{ev} = \begin{cases} 1, & t_d \geq t^{arr} \text{ or } t_d < t^{dep}, \\ 0, & \text{otherwise.} \end{cases}$$

When $A_t^{ev} = 1$, the vehicle is at home and can be charged; when $A_t^{ev} = 0$, the vehicle is away, and the residential charging power must be zero.

1.5.2 EV arrival SoC and Charging Power Constraints

Upon arrival at home, the EV SoC is initialized to the arrival SoC:

$$SOC_{i,t}^{ev} = SOC_{arr}^{ev} = 0.30, \quad t_d = t^{arr}.$$

Thereafter, until the EV departs, its SoC can only increase through residential charging. The EV charging power must satisfy both the connection status and the charger power limit:

$$0 \leq P_{i,t}^{ev} \leq A_t^{ev} P_i^{ev, \max}.$$

Furthermore, the EV SoC must not exceed its upper limit. Expressing the remaining SoC headroom as an equivalent charging power yields an additional upper bound determined by the remaining battery capacity:

$$P_{i,t}^{ev} \leq \frac{(SOC_{\max}^{ev} - SOC_{i,t}^{ev})C_i^{ev}}{\Delta t \eta_{ev}}.$$

Therefore, the physically feasible upper bound on the EV charging power can be written as

$$0 \leq P_{i,t}^{ev} \leq \min \left(A_t^{ev} P_i^{ev, \max}, \frac{(SOC_{\max}^{ev} - SOC_{i,t}^{ev})C_i^{ev}}{\Delta t \eta_{ev}} \right).$$

When the EV is not connected, i.e., $A_t^{ev} = 0$, the above formulation automatically yields $P_{i,t}^{ev} = 0$. As the EV approaches $SOC_{\max}^{ev}$, the SoC headroom constraint progressively limits the charging power, thereby preventing the battery from being charged beyond its physical capacity.

1.5.3 EV SoC Status

The change of the EV SoC is determined by the actual charging power:

$$SOC_{i,t+1}^{ev} = SOC_{i,t}^{ev} + \frac{P_{i,t}^{ev} \Delta t \eta_{ev}}{C_i^{ev}}.$$

Since vehicle-to-grid operation is not considered in this thesis, $P_{i,t}^{ev} \geq 0$. Consequently, the EV SoC is monotonically non-decreasing throughout the connection window. During time steps when the EV is not connected and no arrival SoC reset occurs, the EV does not exchange energy with the residential charging system. At all time steps, the EV SoC must satisfy the following physical constraints:

$$SOC_{\min}^{ev} \leq SOC_{i,t}^{ev} \leq SOC_{\max}^{ev}.$$

1.5.4 EV Departure SoC Requirement

Another physical requirement for the EV is to have sufficient charge when departing. The default departure target SoC is set to

$$SOC_{req}^{ev} = 0.90.$$

Therefore, for each prosumer i , the SoC requirement at departure time is

$$SOC_{i,t^{dep}}^{ev} \geq SOC_{req}^{ev}.$$

Since the connection window spans midnight, this constraint links the charging decisions made after the EV arrives home in the evening to its availability for use on the following morning. Equivalently, the minimum increase in battery energy required to charge the EV from the arrival SoC to the required departure SoC is

$$\Delta E_i^{ev,req} = \max(0, SOC_{req}^{ev} - SOC_{arr}^{ev}) C_i^{ev}.$$

Considering the charging efficiency, the minimum amount of energy that must be supplied from the grid is

$$E_i^{ev,grid,req} = \frac{\Delta E_i^{ev,req}}{\eta_{ev}}.$$

Under the default parameter settings, charging the EV from 0.30 to 0.90 requires an increase of $0.60 \times 60 = 36$ kWh in the battery energy. Accounting for the charging efficiency $\eta_{ev} = 0.95$, the corresponding energy that must be drawn from the grid is approximately

$$\frac{36}{0.95} \approx 37.9 \text{ kWh}.$$

1.6 Electricity Price Model

The electricity price data are obtained from

`datasets/prosumer/price.csv`.

Let p_t denote the wholesale electricity price, expressed in EUR/kWh. The electricity import price used in both the environment and the optimizer is defined as

$$\tilde{p}_t = p_t + \mu^{imp},$$

where μ^{imp} denotes the electricity import price markup. In the current default configuration,

$$\mu^{imp} = 0.$$

Therefore, under the default setting $\tilde{p}_t = p_t$

1.7 Net Load, Power Flow, and Grid Safety

After integrating the load, PV, battery, and EV contributions, the net load injected by prosumer i into the distribution grid at time step t is given by

$$P_{i,t}^{net} = L_{i,t} - G_{i,t}^{use} + P_{i,t}^b + P_{i,t}^{ev}.$$

As mentioned before, when $P_{i,t}^{net} \geq 0$, the prosumer draws power from the grid; when $P_{i,t}^{net} < 0$, it injects power into the grid.

In the pandapower power flow model, positive net load is written to the load table, and negative net load is written to the sgen table:

$$P_{i,t}^{load} = \frac{\max(P_{i,t}^{net}, 0)}{1000},$$

$$P_{i,t}^{sgen} = \frac{\max(-P_{i,t}^{net}, 0)}{1000}.$$

The factor 1000 is used to convert kW to MW. Subsequently, the environment performs AC power flow calculations to obtain the voltages at each bus, line loading rates, and transformer loading rates. In other words, grid safety is not determined by simple algebraic power balance but is jointly decided by the distribution grid topology, line impedances, transformer parameters, and the net injections from each prosumer.

1.7.1 Voltage Constraints

Let $V_{b,t}$ denotes the voltage magnitude at bus b at time step t , expressed in per unit. The voltage safety range is given by

$$V_{\min} = 0.95, \quad V_{\max} = 1.05.$$

Therefore, all buses should satisfy

$$V_{\min} \leq V_{b,t} \leq V_{\max}, \quad b \in \mathcal{B}^{grid}.$$

For any bus b , the voltage violation can be written as

$$v_{b,t} = \max(0, V_{\min} - V_{b,t}) + \max(0, V_{b,t} - V_{\max}).$$

The total voltage violation across the network is measured as

$$\psi_t^V = \sum_{b \in \mathcal{B}^{grid}} v_{b,t}^2.$$

When all bus voltages are within the range $[0.95, 1.05]$, $v_{b,t} = 0$ and $\psi_t^V = 0$. If certain buses fall below or exceed the safety limits due to high load, centralized EV charging, or high PV generation, then ψ_t^V will become positive.

1.7.2 Line Loading Constraints

The safety of lines is measured by their loading percentage. Let $\lambda_{\ell,t}$ denote the loading percentage of line ℓ at time step t . Physically, $\lambda_{\ell,t}$ represents the ratio of the line's apparent power or current to its rated capacity. The current safety limit is

$$\lambda_{\max}^{line} = 100.$$

Therefore, all lines should satisfy

$$\lambda_{\ell,t} \leq \lambda_{\max}^{line}, \quad \ell \in \mathcal{L}.$$

The line violation measure is expressed as

$$\psi_t^{line} = \sum_{\ell} \left(\frac{\max(0, \lambda_{\ell,t} - \lambda_{\max}^{line})}{100} \right)^2,$$

where dividing by 100 is to convert the percentage violation into a relative measure. When all line loadings do not exceed 100%, $\psi_t^{line} = 0$.

1.7.3 Transformer Loading Constraints

Transformer safety is similarly measured by the loading percentage. Let $\lambda_{m,t}^{tr}$ denote the loading percentage of transformer m at time step t . The current safety limit for transformer loading is

$$\lambda_{\max}^{tr} = 100.$$

Therefore, all transformers should satisfy

$$\lambda_{m,t}^{tr} \leq \lambda_{\max}^{tr}, \quad m \in \mathcal{M}^{tr}.$$

The corresponding transformer violation measure is

$$\psi_t^{tr} = \sum_m \left(\frac{\max(0, \lambda_{m,t}^{tr} - \lambda_{\max}^{tr})}{100} \right)^2.$$

Transformer constraints are particularly sensitive to the system's total net load. When multiple prosumers charge their EVs or batteries simultaneously, the power demand at the feeder head may increase, thereby raising transformer loading; when PV generation is high and local load is low, reverse power flow may also affect transformer and line states.

In summary, grid safety constraints can be summarized as

$$\begin{aligned} V_{\min} &\leq V_{b,t} \leq V_{\max}, & b &\in \mathcal{B}^{grid}, \\ \lambda_{\ell,t} &\leq \lambda_{\max}^{line}, & \ell &\in \mathcal{L}, \\ \lambda_{m,t}^{tr} &\leq \lambda_{\max}^{tr}, & m &\in \mathcal{M}^{tr}. \end{aligned}$$

These safety metrics do not alter the physical power flow results themselves, but are used to evaluate whether given control actions lead to voltage violations, line overloads, or transformer overloads.

1.8 Optimization Objective

The objective of this study is to minimize the total electricity cost of the system while satisfying both the device operational constraints and the distribution network constraints. For prosumer i , the economic cost associated with the stationary battery at time step t is given by

$$C_{i,t}^b = \tilde{p}_t \Delta t P_{i,t}^b.$$

Since $P_{i,t}^b > 0$ denotes battery charging, $C_{i,t}^b$ represents a positive charging cost. Conversely, when $P_{i,t}^b < 0$, the battery is discharging, resulting in a negative value of $C_{i,t}^b$,

which corresponds to a reduction in the electricity purchasing cost or, equivalently, revenue from battery discharge.

The EV charging cost is given by

$$C_{i,t}^{ev} = \tilde{p}_t \Delta t P_{i,t}^{ev}.$$

In the current model, PV curtailment does not have an explicit economic price, thus

$$C_{i,t}^{pv} = 0.$$

Therefore, the total economic cost of the single-step system is

$$C_t = \sum_{i=1}^N \left(C_{i,t}^b + C_{i,t}^{ev} + C_{i,t}^{pv} \right).$$

In an episode, the objective function can be formulated as

$$\begin{aligned} \min_{\{P^b, G^{use}, P^{ev}\}} \quad & \sum_{t=0}^{T-1} \sum_{i=1}^N \tilde{p}_t \Delta t \left(P_{i,t}^b + P_{i,t}^{ev} \right) \\ \text{s.t.} \quad & L_{i,t} = s_i^L \left(L_{i,t}^{hh,raw} + L_{i,t}^{hp,raw} \right), \\ & 0 \leq G_{i,t}^{use} \leq G_{i,t}, \\ & -P_i^{b,max} \leq P_{i,t}^b \leq P_i^{b,max}, \\ & SOC_{min}^b \leq SOC_{i,t}^b \leq SOC_{max}^b, \\ & 0 \leq P_{i,t}^{ev} \leq A_i^{ev} P_i^{ev,max}, \\ & SOC_{min}^{ev} \leq SOC_{i,t}^{ev} \leq SOC_{max}^{ev}, \\ & SOC_{i,t}^{ev,dep} \geq SOC_{req}^{ev}, \\ & P_{i,t}^{net} = L_{i,t} - G_{i,t}^{use} + P_{i,t}^b + P_{i,t}^{ev}, \\ & V_{min} \leq V_{b,t} \leq V_{max}, \\ & \lambda_{\ell,t} \leq \lambda_{max}^{line}, \\ & \lambda_{m,t}^{tr} \leq \lambda_{max}^{tr}. \end{aligned}$$

The problem has a multi-time-step coupling characteristic: the battery SoC and EV SoC make the current actions affect the future feasible region; PV, load, and price exhibit obvious time variations; and the distribution network flow causes different prosumer actions to be coupled through voltage and line loading. Therefore, this study will model the system as a multi-agent decision problem and compare the performance of MADRL, MPC, and baselines in terms of economic efficiency and grid safety.

2 Methodology

This chapter presents the prediction, control, safety correction, and training methods adopted in this thesis. Building upon the system model introduced in Chapter 3, this chapter further describes how these models are implemented in the code. The overall framework consists of an LSTM-based forecasting module, a pandapower-based simulation environment, a multi-agent deep reinforcement learning controller, a safety action projection mechanism, and an EV constraint handling mechanism.

2.1 Overall Framework

The overall workflow can be summarized as follows. First, the historical data are processed into time-series sequences of load demand, PV generation, and electricity prices. The LSTM predictor then generates forecasts over a future prediction horizon. The environment constructs the observation by combining the current system state with the predicted sequences. Based on its own observation, each MADRL actor outputs a control action. The generated actions subsequently undergo local feasibility checks and, if enabled, a safety action projection procedure. The environment then updates the battery SoC, EV SoC, and the net load of each prosumer according to the final control actions. Next, pandapower performs the power flow calculation using the updated net loads. Finally, the environment computes the reward based on the electricity cost, EV charging requirements, battery behavior, and the distribution network operating condition, and returns the reward to the agents for training.

Let x_t denote the external input data at the current time step, including load $L_{i,t}$, PV generation $G_{i,t}$, and electricity price p_t . Let $\hat{x}_{t:t+H-1}$ denote the prediction window of length H . The process within a time step is given by

$$x_t, \hat{x}_{t:t+H-1}, SOC_t^b, SOC_t^{ev} \rightarrow o_t \rightarrow a_t \rightarrow \bar{a}_t \rightarrow s_{t+1}, r_t.$$

where o_t is the observation, a_t is the actor's raw action, $\bar{a}_t$ is the executed action after feasibility checks or projection, s_{t+1} is the environment's next state, and r_t is the immediate reward for each agent.

The overall framework of this work includes the following steps:

1. Read historical household load, PV generation, and electricity price data.

2. Use LSTM model to generate future prediction sequences for electricity price, load, and PV generation.
3. The environment constructs the observation based on the current SoC, EV connection status, current exogenous variables, and the prediction window.
4. Each agent's actor outputs three action components: battery action, PV action, and EV action.
5. The battery action is first mapped to the feasible power range of SoC; in MADRL_PROJECTION, battery power and PV curtailment are further subjected to grid safety projection.
6. The environment updates the battery SoC, EV SoC, PV utilization, and net load based on the final actions.
7. pandapower writes the prosumer net load to the low-voltage distribution network and performs power flow calculation.
8. The environment calculates the reward based on economic terms, EV-related terms, battery-related terms, and grid safety terms.
9. The transition is written to the replay buffer and is used to update the actor and critic networks.

Figure 2.1 illustrates the main workflow of this work.

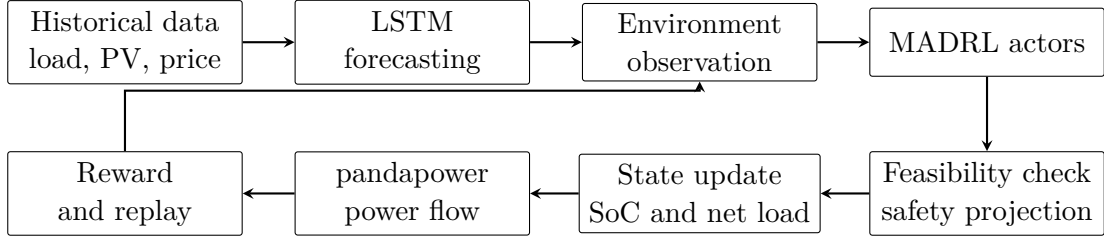


Figure 2.1: The work flow of the MADRL energy management framework.

2.2 LSTM Forecasting

This thesis employs an LSTM-based forecasting module to provide future-horizon information to the controller. The forecasting targets include the electricity price, prosumer load, and PV generation. The prediction results determine the amount of future

information available to the controller and therefore directly influence the scheduling performance of the stationary battery, PV curtailment, and EV charging.

For any scalar sequence y_t , LSTM uses a historical window of length h as input and outputs a predictive window of length H . The current project configuration is

$$h = 192, \quad H = 49.$$

Since the simulation time step is 0.25 h, a historical window of 192 time steps corresponds to the previous 48 hours, while a prediction window of 49 time steps includes the current time step and approximately the following 12 hours. The training sample can be formulated as

$$\begin{aligned} X_k &= (y_{k-h}, y_{k-h+1}, \dots, y_{k-1}), \\ Y_k &= (y_k, y_{k+1}, \dots, y_{k+H-1}). \end{aligned}$$

The LSTM learns the mapping

$$\hat{Y}_k = f_\theta(X_k, \phi_k),$$

where ϕ_k denotes the optional temporal features, such as the hour of the day, the day of the week, or annual cyclic features.

In this work, the electricity price is predicted using a separate LSTM model; the PV generation is predicted using a shared model trained on reference PV sequences; and the load prediction is organized by prosumer and load components.

The predicted sequences used in the environment are

$$\hat{p}_{t:t+H-1}, \quad \hat{L}_{i,t:t+H-1}, \quad \hat{G}_{i,t:t+H-1}.$$

In the perfect forecast mode, these sequences are directly obtained from the ground-truth future values. In the LSTM forecast mode, they are generated by the trained LSTM predictor. Unless otherwise specified, the MADRL controller is trained using the LSTM forecast mode.

2.3 MADRL Observation

Observation is divided into three components: the local observation, the forecast sequence observation, and the safety observation. This design follows the centralized training and decentralized execution (CTDE) paradigm. Specifically, each actor makes decentralized decisions primarily based on its own local state and the forecast horizon, while the critic has access to the joint information of all agents during the training process.

2.3.1 Local Observation

The local observation contains the key physical states of each agent at the current time step:

$$o_{i,t}^{local} = [SOC_{i,t}^b, SOC_{i,t}^{ev}, A_t^{ev}, L_{i,t}, G_{i,t}, p_t].$$

Here, $SOC_{i,t}^b$ denotes the SoC of the stationary battery, $SOC_{i,t}^{ev}$ denotes the EV SoC, A_t^{ev} indicates whether the EV is connected to the residential charger, $L_{i,t}$ represents the current electrical load, $G_{i,t}$ denotes the currently available PV generation, and p_t is the current electricity price.

In the MADRL actor, the local information is further transformed into `madr1_local`. This feature includes intraday and annual cyclic time features, normalized battery SoC, normalized EV SoC, and EV availability indicators:

$$o_{i,t}^{madr1} = [\sin(2\pi h_t/24), \cos(2\pi h_t/24), \sin(2\pi d_t/365.25), \cos(2\pi d_t/365.25), \bar{SOC}_{i,t}^b, \bar{SOC}_{i,t}^{ev}, A_t^{ev}].$$

This representation enables the actor not only to perceive the current states of the devices but also to identify the current temporal position within both the daily and annual cycles.

2.3.2 Forecast Sequence Observation

The forecast sequence observation describes the future trajectories of the electricity price, load demand, and PV generation. In the environment, the **sequence** can be expressed as

$$o_{i,t}^{seq} = [\hat{p}_{t:t+H-1}, \hat{L}_{i,t:t+H-1}, \hat{G}_{i,t:t+H-1}].$$

In practice, the MADRL actor uses normalized or relative representations of these forecast sequences:

$$\bar{p}_{t:t+H-1}, \quad \Delta\bar{p}_{t:t+H-1}, \quad \bar{L}_{i,t:t+H-1}, \quad \bar{G}_{i,t:t+H-1}.$$

where $\bar{p}$ denotes the relative electricity price within the prediction horizon, $\Delta\bar{p}$ represents the price variation intensity over the prediction window, and $\bar{L}$ and $\bar{G}$ denote the normalized load and PV generation forecasts, respectively. Physically, these features enable the actor to schedule the stationary battery and EV charging according to anticipated periods of high or low electricity prices, while also determining whether battery capacity should be reserved or utilized based on the forecasted PV generation.

2.3.3 Safety Observation

The safety observation is used for action feasibility checking and the safety projection procedure. It contains the physical parameters required to determine the device operating

limits and perform the grid safety projection:

$$o_{i,t}^{safe} = [SOC_{i,t}^b, SOC_{i,t}^{ev}, A_t^{ev}, L_{i,t}, G_{i,t}, C_i^b, P_i^{b,max}, C_i^{ev}, P_i^{ev,max}].$$

Here, the battery capacity, battery maximum charging and discharging power, EV battery capacity, and EV maximum charging power are not merely static configuration parameters; they also directly determine the physically feasible action space. For example, the battery SoC together with $P_i^{b,max}$ determines the maximum amount of energy that can be charged into or discharged from the battery at the current time step. Similarly, the EV connection status together with $P_i^{ev,max}$ determines whether the EV is currently available for charging and, if so, the maximum allowable charging power.

2.4 MADRL Action Design

Each agent's actor outputs a three-dimensional action vector:

$$a_{i,t} = [a_{i,t}^b, a_{i,t}^{pv}, a_{i,t}^{ev}] \in [-1, 1]^3.$$

The three action components correspond to the stationary battery, PV utilization, and EV charging, respectively.

2.4.1 Battery Action Mapping

The raw battery action generated by the actor is not executed directly as the physical battery power. Instead, it is first mapped to the feasible power range determined by the current battery SoC. Given the current SoC, the battery cannot be charged beyond SOC_{max}^b or discharged below SOC_{min}^b within a single time step. Therefore, the feasible battery power limits at the current time step are given by

$$P_{i,t}^{b,low} = -\min\left(P_i^{b,max}, \frac{(SOC_{i,t}^b - SOC_{min}^b)C_i^b\eta_b}{\Delta t}\right),$$

$$P_{i,t}^{b,up} = \min\left(P_i^{b,max}, \frac{(SOC_{max}^b - SOC_{i,t}^b)C_i^b}{\eta_b\Delta t}\right).$$

The raw action $a_{i,t}^b \in [-1, 1]$ is linearly mapped to this interval:

$$P_{i,t}^b = P_{i,t}^{b,low} + \frac{a_{i,t}^b + 1}{2}(P_{i,t}^{b,up} - P_{i,t}^{b,low}).$$

This ensures that even if the actor outputs a boundary value, the executed action does not directly cause the battery SoC to violate its limits.

2.4.2 PV Action Mapping

The PV action represents the PV utilization factor. It is defined as

$$u_{i,t}^{pv} = \min \left(1, \max \left(0, \frac{a_{i,t}^{pv} + 1}{2} \right) \right).$$

The corresponding utilized PV generation and curtailed PV power are given by

$$G_{i,t}^{use} = u_{i,t}^{pv} G_{i,t},$$

$$G_{i,t}^{curt} = G_{i,t} - G_{i,t}^{use}.$$

When $a_{i,t}^{pv} = 1$, the agent uses all available PV; when $a_{i,t}^{pv} = -1$, the PV is completely curtailed. The physical meaning of the PV action is to reduce reverse power flow or voltage violation risks through controllable curtailment.

2.4.3 EV Action Mapping

The EV action represents the utilization factor of the residential charger. It is defined as

$$u_{i,t}^{ev} = \min \left(1, \max \left(0, \frac{a_{i,t}^{ev} + 1}{2} \right) \right).$$

Then, EV charging power requested by the actor is

$$P_{i,t}^{ev,rl} = A_t^{ev} u_{i,t}^{ev} P_i^{ev,max}.$$

If the EV is not connected, $A_t^{ev} = 0$, the EV charging power is always zero regardless of the action produced by the actor. Subsequently, the environment determines the final executed charging power $P_{i,t}^{ev}$ by considering the EV SoC headroom as well as the hard projection or emergency charging mechanism, if enabled.

2.5 Reward Function Design

The reward function combines the economic objective, EV mobility requirements, battery behavior, and distribution network safety. For each agent i , the total reward consists of multiple components. Positive terms represent benefits or incentives, whereas negative terms represent costs or constraint violations.

2.5.1 EV Charging Cost

EV charging draws electrical energy from either the grid or local energy resources. Its economic cost is determined by the actual charging power and the electricity price. Let

$$\tilde{p}_t = p_t + \mu^{imp}$$

denote the electricity price. The basic EV charging cost is then given by

$$C_{i,t}^{ev} = w_{cost}^{ev} P_{i,t}^{ev} \tilde{p}_t \Delta t.$$

The corresponding reward term is

$$r_{i,t}^{ev,cost} = -C_{i,t}^{ev}.$$

EV charging during high-price periods incurs higher costs, while charging during low-price periods is more economical.

2.5.2 EV Costweight Modelling

Experimental results show that using only the EV charging cost as the reward encourages overly conservative EV charging behavior, even during periods of low electricity prices. The parameter w_{cost}^{ev} is therefore introduced to adjust the relative importance of the EV charging cost in the MADRL reward function. In the current implementation, the soft and hard modes use the configuration parameter `ev_charging_cost_weight`, whereas the emergency mode uses a fixed value of 1:

$$w_{cost}^{ev} = \begin{cases} \text{ev_charging_cost_weight}, & \text{mode} \in \{\text{soft}, \text{hard}\}, \\ 1, & \text{mode} = \text{emergency}. \end{cases}$$

When $w_{cost}^{ev} > 1$, the agent places greater emphasis on avoiding costly EV charging during periods of high electricity prices. When $w_{cost}^{ev} = 1$, the EV charging cost is calculated directly according to the actual electricity price.

2.5.3 EV Departure Penalty

The EV departure requirement represents a physical mobility constraint, requiring the EV to reach the target SoC before departure. In the reward function, the departure penalty converts this requirement into a learning signal under the soft mode. The SoC deficit at the departure time step is defined as

$$g_{i,t}^{dep} = \max(0, SOC_{req}^{ev} - SOC_{i,t+1}^{ev}), \quad t_d = t^{dep}.$$

The corresponding penalty is

$$L_{i,t}^{ev,dep} = w_{dep}^{ev} (g_{i,t}^{dep})^2.$$

This penalty is activated only at the departure time step. Physically, it indicates that if the EV has not reached the required SoC by the time of departure, the user's requirement has not been satisfied. Consequently, the reward function assigns a larger negative feedback to discourage such behavior.

2.5.4 EV Progress Penalty

Experimental results indicate that applying the departure penalty only at the departure time step is still insufficient to consistently achieve the required departure SoC. To alleviate this issue, an additional progress penalty can be enabled in the soft EV experiments. This penalty provides a linear reference trajectory from the arrival SoC to the required departure SoC throughout the EV connection window. Let $\alpha_t \in [0, 1]$ denote the normalized progress within the current connection window. Then,

$$SOC_t^{ev,prog} = SOC_{arr}^{ev} + \alpha_t (SOC_{req}^{ev} - SOC_{arr}^{ev}).$$

if the EV SoC falls below this reference trajectory, a penalty is applied to encourage the agent to gradually approach the departure target during the night connection window. The progress gap can be expressed as

$$g_{i,t}^{prog} = \max(0, SOC_t^{ev,prog} - SOC_{i,t+1}^{ev}),$$

The corresponding penalty is

$$L_{i,t}^{ev,prog} = w_{prog}^{ev} (g_{i,t}^{prog})^2.$$

This term encourages the EV to gradually approach the required departure SoC throughout the overnight connection window, rather than postponing the entire charging demand until the final time steps before departure.

2.5.5 Price-Aware Penalty

During the experiments, it was further observed that increasing the EV costweight introduces competition between the EV reward and the battery reward, leading to unsatisfactory overall performance. Moreover, the EV is still unable to completely avoid charging during periods of high electricity prices. To further encourage the EV to avoid expensive charging periods, a price-aware penalty is introduced in this thesis. First, a quantile threshold is computed from the electricity price profile within the current episode:

$$p^{thr} = Q_q(\{p_\tau\}),$$

where q denotes `ev_price_aware_threshold_quantile`, whose used value is 0.70. When the current electricity price exceeds this threshold, the normalized excess electricity price is defined as

$$\delta_t^p = \frac{\max(0, \tilde{p}_t - p^{thr})}{\max(p^{thr}, 10^{-6})}.$$

The price-aware penalty is then defined as

$$L_{i,t}^{ev,price} = w_{price}^{ev} P_{i,t}^{ev} \delta_t^p \Delta t.$$

When the electricity price does not exceed the threshold, this penalty is zero. When the electricity price exceeds the threshold and the EV is charging, this term becomes a positive penalty, encouraging the agent to shift flexible EV charging demand to periods with relatively lower electricity prices.

2.5.6 Battery Reward and Regularization

The battery reward consists of four components: the economic benefit of battery charging and discharging, the boundary action penalty, the SoC regularization term, and the throughput bonus applied during the early stage of training. The instantaneous economic benefit of the battery is given by

$$r_{i,t}^{bat} = [\max(-P_{i,t}^b, 0) - \max(P_{i,t}^b, 0)] \tilde{p}_t \Delta t.$$

When the battery is already close to its SoC limits but continues to move toward the corresponding boundary, an action boundary penalty is imposed:

$$L_{i,t}^{bat,act} = w_{act}^{bat} |P_{i,t}^b| I_{i,t}^{bd}.$$

where $I_{i,t}^{bd} = 1$ indicates that the battery is discharging near a low SoC limit or charging near a high SoC limit; otherwise, $I_{i,t}^{bd} = 0$. Additionally, the SoC regularization term encourages the battery to stay away from the hard limits:

$$L_{i,t}^{bat,soc} = w_{soc}^{bat} [\max(0, SOC_{low}^{soft} - SOC_{i,t}^b)^2 + \max(0, SOC_{i,t}^b - SOC_{high}^{soft})^2].$$

The throughput bonus for battery applied during the early stage of training is given by

$$r_{i,t}^{bat,thr} = w_t^{thr} |P_{i,t}^b| \Delta t.$$

This term gradually decays with training progress, aiming to prevent the actor from learning overly conservative strategies that avoid using the battery entirely in the early stages of training.

2.5.7 Grid Safety Penalty

The grid safety penalty is derived from the results of the pandapower power flow calculation and consists of the three categories introduced previously: voltage violations, line loading violations, and transformer loading violations, as mentioned before. Let ψ_t^V , ψ_t^{line} , and ψ_t^{tr} denote the raw violation measures for voltage, line loading, and transformer loading, respectively. The corresponding penalty terms are

$$L_{i,t}^V = Nw_V\psi_t^V\omega_{i,t}^V,$$

$$L_{i,t}^{line} = w_{line}\psi_t^{line},$$

$$L_{i,t}^{tr} = w_{tr}\psi_t^{tr}.$$

where $\omega_{i,t}^V$ denotes the voltage penalty weight based on the degree of voltage violation for each agent bus; line and transformer penalties are currently the same for all agents. This design ensures that agents not only focus on their own economic benefits but also perceive the impact of their actions on the safety of the distribution grid.

2.5.8 Total Reward

The total reward for each agent can be written as

$$r_{i,t} = r_{i,t}^{bat} - C_{i,t}^{ev} - L_{i,t}^{bat,act} - L_{i,t}^{bat,soc} - L_{i,t}^{ev,dep} - L_{i,t}^{ev,proj} - L_{i,t}^{ev,price} - L_{i,t}^{ev,proj} - L_{i,t}^{ev,eme} + r_{i,t}^{bat,thr} - L_{i,t}^V - L_{i,t}^{line} - L_{i,t}^{tr}.$$

Here, $L_{i,t}^{ev,proj}$ and $L_{i,t}^{ev,eme}$ denote the diagnostic penalty terms introduced by the hard projection and emergency charging mechanisms, respectively, which will be discussed in the following sections. Overall, the reward function is designed to simultaneously encourage cost-efficient operation, satisfy the EV mobility requirements, promote appropriate battery utilization, and prevent violations of the distribution network safety constraints.

2.6 Safety Projection for Grid Constraints

The reward penalty represents a soft constraint: it allows the agent to take unsafe actions but penalizes such behavior through the reward function. In contrast, the safety projection acts as a hard correction mechanism applied before action execution. It directly modifies the control actions before they are executed by the environment, thereby moving them closer to the grid-safe feasible region. In the proposed MADRL_PROJECTION framework, the safety projection is employed as a complement to the reward penalty mechanism.

The decision variables of the projection are not the complete action vector but only the battery power and the PV curtailment:

$$x_t = [P_{1,t}^b, \dots, P_{N,t}^b, G_{1,t}^{curt}, \dots, G_{N,t}^{curt}].$$

Given the current load, PV generation, and EV charging power, the base net load is

$$P_{i,t}^{base} = L_{i,t} - G_{i,t} + P_{i,t}^{ev}.$$

The safety projector uses pandapower to compute finite-difference sensitivities of the bus voltages and line loading with respect to the agents' net power injections around the zero-injection operating point. The corresponding linearized relationships can be written as

$$\begin{aligned} V_t &\approx V^{base} + S^V \Delta P_t, \\ \lambda_t^{line} &\approx \lambda^{base} + S^{line} \Delta P_t. \end{aligned}$$

where ΔP_t is determined by the battery power and PV curtailment. The projection goal is to satisfy the linearized safety constraints with respect to voltage, line, and transformer limits while making minimal changes to the original actions.

Each linearized constraint can be written in half-space form

$$r_j^T x \leq b_j.$$

If the current solution x violates the corresponding constraint, it is projected sequentially onto each constraint hyperplane as

$$x \leftarrow x - \frac{\max(0, r_j^T x - b_j)}{\|r_j\|_2^2} r_j.$$

After the projection step, the battery power is clipped again to the locally feasible SoC range, while the PV curtailment is clipped to

$$0 \leq G_{i,t}^{curt} \leq G_{i,t}.$$

This procedure is repeated for several iterations. Physically, the projection mitigates voltage and line loading violations by reducing battery charging, increasing battery discharging, or increasing PV curtailment. Since EV charging is primarily driven by the user's mobility requirements, this thesis treats the EV constraints separately from the grid safety projection.

2.7 EV Departure Requirement Handling

This thesis implements three approaches for handling the EV departure requirement: the soft, hard, and emergency charging strategies. All three approaches employ the same physical EV model, but they differ in terms of action execution and reward feedback.

2.7.1 Soft Mode

Soft mode directly executes the EV charging power requested by the actor, and guides policy learning through departure penalties, progress penalties, or price-aware penalties in the reward function. The executed power is

$$P_{i,t}^{ev} = P_{i,t}^{ev,rl}$$

and is subject to constraints from EV connection, charger power, and SoC headroom. The advantage of soft mode is that it allows for high policy freedom, enabling agents to learn price responsiveness; however, its disadvantage is that the departure requirement mainly relies on reward learning, and if the reward signal is sparse, it may result in insufficient SoC at home.

2.7.2 Hard Mode

In the hard mode, the execution layer checks whether the current EV charging action is sufficient to preserve the feasibility of meeting the required departure SoC. Let K_t^{future} denote the number of remaining time steps during which the EV will remain connected before departure. The maximum achievable increase in SoC over the remaining connection window is

$$\Delta SOC_{i,t}^{future,max} = \frac{K_t^{future} P_i^{ev,max} \Delta t \eta_{ev}}{C_i^{ev}}.$$

If the EV is not charged sufficiently at the current time step, it may become impossible to reach SOC_{req}^{ev} before departure, even if the charger subsequently operates at its maximum power. Therefore, the hard mode computes the minimum required charging power at the current time step as

$$P_{i,t}^{ev,req} = \max \left(0, \frac{(SOC_{req}^{ev} - \Delta SOC_{i,t}^{future,max} - SOC_{i,t}^{ev}) C_i^{ev}}{\Delta t \eta_{ev}} \right).$$

The finally executed charging power is then determined by

$$P_{i,t}^{ev} = \min \left(P_i^{ev,max}, \max(P_{i,t}^{ev,rl}, P_{i,t}^{ev,req}) \right).$$

The physical meaning is that the control policy is free to select a charging power higher than the minimum required value, but it is not allowed to choose a charging power below the minimum necessary level.

2.7.3 Emergency Charging

Emergency charging is a remedial mechanism designed to compensate for the potential SoC deficiency that may occur under the soft mode. It is activated only within an emergency window immediately before the EV departure time. Let the emergency window consist of K^{eme} time steps. When the EV enters the emergency window and its SoC remains below the required target, the system computes the required charging power based on the remaining SoC deficit:

$$E_{i,t}^{need} = \max(0, SOC_{req}^{ev} - SOC_{i,t}^{ev})C_i^{ev},$$

$$P_{i,t}^{ev,eme} = \min\left(P_i^{ev,max}, \frac{E_{i,t}^{need}}{K_t^{rem}\Delta t\eta_{ev}}\right).$$

The finally executed power is then determined by

$$P_{i,t}^{ev} = \max(P_{i,t}^{ev,rl}, P_{i,t}^{ev,eme}).$$

Unlike the hard mode, the emergency mode does not continuously evaluate future feasibility throughout the entire EV connection window. Instead, it performs remedial charging only when the departure time is approaching. Therefore, the emergency mode is simpler to implement, but it may result in a large concentration of charging power during the final stage before departure.

2.8 MADRL Training Algorithm

This thesis adopts a multi-agent actor-critic framework based on the centralized training and decentralized execution paradigm. Each agent is equipped with an independent actor:

$$a_{i,t} = \pi_i(o_{i,t}; \theta_i),$$

where $o_{i,t}$ denotes the actor observation of agent i . During training, a centralized critic is employed, which receives the observations and the actions of all agents:

$$Q_i(o_{1:N,t}, a_{1:N,t}; \phi_i).$$

To reduce Q-value overestimation, the code employs twin critics for each agent:

$$Q_{i,1}, \quad Q_{i,2}.$$

Replay buffer saves transition:

$$(o_t, a_t, R_t^{(n)}, o_{t+n}, d_t),$$

where $R_t^{(n)}$ is the n-step return:

$$R_{i,t}^{(n)} = \sum_{k=0}^{n-1} \gamma^k r_{i,t+k}.$$

Here, $n = 48$ corresponds to an accumulated reward over approximately 12 hours. The critic target is given by

$$y_{i,t} = R_{i,t}^{(n)} + \gamma^n (1 - d_t) \min(Q'_{i,1}(o_{t+n}, a'_{t+n}), Q'_{i,2}(o_{t+n}, a'_{t+n})).$$

Target actions are added with TD3-style target policy smoothing noise:

$$a'_{t+n} = \text{guard}(\pi'(o_{t+n}) + \epsilon).$$

where **guard** denotes the local SoC feasibility check and optional safety projection.

Critic loss is

$$L_i^Q = (Q_{i,1}(o_t, a_t) - y_{i,t})^2 + (Q_{i,2}(o_t, a_t) - y_{i,t})^2.$$

During the actor update, only the i -th action in the joint action vector is replaced:

$$a_{i,t}^{pol} = \text{guard}(\pi_i(o_{i,t})),$$

$$L_i^\pi = -Q_{i,1}(o_t, a_{1,t}, \dots, a_{i,t}^{pol}, \dots, a_{N,t}).$$

Target network 采用软更新:

$$\theta'_i \leftarrow (1 - \tau)\theta'_i + \tau\theta_i,$$

$$\phi'_i \leftarrow (1 - \tau)\phi'_i + \tau\phi_i.$$

During training, both exploration noise and feasible random exploration are employed. The former adds random perturbations to the actor output, while the latter samples locally feasible battery actions with a certain probability during the early stage of training, thereby helping the agent explore the feasible battery SoC operating region.

2.8.1 Training Pseudocode

The overall MADRL training procedure adopted in this thesis can be summarized by the following pseudocode:

1. Initialize N actors, N twin critics, the corresponding target networks, and the replay buffer.

2. Reset the parallel training environments and obtain the initial observations.
3. For each environment step:
 - a) Each actor generates a raw action based on its current observation.
 - b) Add exploration noise and perform the local feasibility check.
 - c) If MADRL_PROJECTION is enabled, perform the safety projection.
 - d) Execute the actions in the environment, update the battery SoC, EV SoC, and net load, and perform the power flow calculation using pandapower.
 - e) The environment returns the reward, the next observation, and the done flag.
 - f) Store the transition in the replay buffer and compute the n -step return.
 - g) Once the replay buffer contains sufficient samples, draw a mini-batch to update the critics.
 - h) Update the actors and target networks every predefined number of training steps.
4. Repeat the above procedure until the specified number of training episodes is reached.

The algorithm is decentralized during execution: each actor only needs its own local observation and prediction sequence to output an action; however, during training, the critic uses joint observations and joint actions to learn the coupling effects between agents through the power grid flow.

2.9 Baseline Controllers

To evaluate the performance of the proposed MADRL approach, this thesis also implements optimization-based baseline controllers, including Local MPC and ADMM-MPC.

2.9.1 Local MPC

Local MPC solves an independent receding-horizon optimization problem for each prosumer. It neither considers the actions of the other prosumers nor explicitly enforces network-wide voltage or line loading constraints. Instead, it optimizes its own battery operation, PV curtailment, and EV charging schedule over the prediction horizon. For prosumer i , the primary decision variables of the Local MPC formulation include

$$c_{i,\tau}, d_{i,\tau}, G_{i,\tau}^{curt}, P_{i,\tau}^{ev}, E_{i,\tau}^b, E_{i,\tau}^{ev}.$$

where c and d denote the battery charging power and discharging power, respectively, with

$$P_{i,\tau}^b = c_{i,\tau} - d_{i,\tau}.$$

Function for the battery is:

$$E_{i,\tau+1}^b = E_{i,\tau}^b + \eta_b c_{i,\tau} \Delta t - \frac{d_{i,\tau} \Delta t}{\eta_b}.$$

Function for the EV is:

$$E_{i,\tau+1}^{ev} = E_{i,\tau}^{ev} + \eta_{ev} P_{i,\tau}^{ev} \Delta t.$$

Local net load is:

$$P_{i,\tau}^{net} = L_{i,\tau} - G_{i,\tau} + G_{i,\tau}^{curt} + c_{i,\tau} - d_{i,\tau} + P_{i,\tau}^{ev}.$$

Local MPC objective function mainly minimizes the battery charging cost, discharging benefit, and EV charging cost:

$$\min \sum_{\tau} \tilde{p}_{\tau} \Delta t (c_{i,\tau} - d_{i,\tau} + P_{i,\tau}^{ev}) + \epsilon_b \Delta t (c_{i,\tau} + d_{i,\tau}) + w_{dep}^{ev} g_i^2.$$

where ϵ_b is a small throughput tie-breaker, used to avoid unnecessary simultaneous charging and discharging or equivalent solutions; g_i is the EV departure gap. The advantage of Local MPC is that it has a small problem size and fast solution time; its disadvantage is the lack of grid coordination across agents.

2.9.2 ADMM-MPC

ADMM-MPC introduces distributed coordination on top of Local MPC, aiming to constrain the impact of aggregated net load on transformers. Each agent still solves its local subproblem, but additionally introduces local net load contribution

$$q_{i,\tau} = \alpha_{i,\tau} P_{i,\tau}^{net}.$$

In the current implementation, $\alpha_{i,\tau} = 1$. The global coordination objective is to ensure that all agents' contributions satisfy the transformer aggregation bounds:

$$-\bar{P}^{tr} \leq \sum_i q_{i,\tau} \leq \bar{P}^{tr}.$$

ADMM adds a quadratic penalty term to the local objective function :

$$J_i^{ADMM} = J_i^{local} + \frac{\rho}{2} \|\alpha_i P_i^{net} - z_i + u_i\|_2^2.$$

where z_i is the coordination variable, u_i is the scaled dual variable, and ρ is the ADMM penalty parameter. The solution process iterates between local optimization and aggregation projection:

$$\begin{aligned} z &\leftarrow \Pi_{[-p_{tr}, p_{tr}]}(q + u), \\ u &\leftarrow u + q - z. \end{aligned}$$

ADMM-MPC preserves the local optimization structure of each agent while reducing the risk of transformer overloading through coordination based on coupling constraints. Compared with Local MPC, it provides a certain degree of system-level coordination. Compared with a centralized MISOCP formulation, its computational structure is more decentralized.

2.9.3 Rule-Based Controllers

In addition to the two approaches described above, this thesis also implements two rule-based EV controllers as baseline methods. These controllers provide simple reference strategies that do not rely on either learning algorithms or optimization solvers. The two controllers are `rule_based_ev_no_battery` and `rule_based_ev_with_battery`.

The first controller is the no-battery rule-based controller. This controller only manages EV charging, without using the stationary battery or performing PV curtailment. For each prosumer i , the battery power and PV curtailment are fixed to 0. The EV charging rule depends only on the EV connection status and whether the EV SoC has reached its upper limit. If the EV is not connected, no charging is performed. If the EV is connected and $SOC_{i,t}^{ev} < SOC_{\max}^{ev}$, the EV is charged at the maximum charging power:

$$P_{i,t}^{ev} = \begin{cases} P_i^{ev,\max}, & A_t^{ev} = 1 \text{ and } SOC_{i,t}^{ev} < SOC_{\max}^{ev}, \\ 0, & \text{otherwise.} \end{cases}$$

This controller represents the simplest possible charging behavior: once the vehicle arrives home, it immediately charges at the maximum power of the residential charger until the battery is fully charged. It does not consider the electricity price or the state of the distribution network and therefore serves as a baseline for EV charging. Since no stationary battery is available, the network security mainly depends on the maximum EV charging power constraint.

The second controller is the with-battery rule-based controller. It extends the no-battery EV charging strategy by incorporating a price-threshold rule for the stationary battery. The EV charging strategy remains identical, i.e.,

$$P_{i,t}^{ev} = \begin{cases} P_i^{ev,\max}, & A_t^{ev} = 1 \text{ and } SOC_{i,t}^{ev} < SOC_{\max}^{ev}, \\ 0, & \text{otherwise.} \end{cases}$$

The difference is the battery operation, which is determined according to the current electricity import price and the quantiles of the training electricity price distribution. Let $\tilde{p}_t$ denote the current electricity import price, and let p^{low} and p^{high} denote the lower and upper quantile thresholds of the electricity price series used during training:

$$\begin{aligned} p^{low} &= Q_{q_{low}}(\tilde{p}^{train}), \\ p^{high} &= Q_{q_{high}}(\tilde{p}^{train}). \end{aligned}$$

Based on these two thresholds, the battery power is determined as

$$P_{i,t}^b = \begin{cases} \alpha_c P_i^{b,\max}, & \tilde{p}_t \leq p^{low}, \\ -\alpha_d P_i^{b,\max}, & \tilde{p}_t \geq p^{high}, \\ 0, & p^{low} < \tilde{p}_t < p^{high}. \end{cases}$$

where $\alpha_c \in [0, 1]$ and $\alpha_d \in [0, 1]$ denote the charging and discharging ratios, respectively. In the current implementation, both parameters are set to 1. Consequently, the battery is commanded to charge at its maximum power during low-price periods, discharge at its maximum power during high-price periods, and remain idle when the electricity price falls between the two thresholds. The resulting battery action is still processed by the battery SoC feasibility mapping in the environment. Therefore, even if the rule requests maximum charging or discharging power, the executed battery power will never violate the SoC limits $[SOC_{\min}^b, SOC_{\max}^b]$.

The physical meaning of this controller is : electrical energy is stored during periods of low electricity prices and released during periods of high electricity prices. The controller requires neither future forecasts nor the solution of an optimization problem, resulting in a very low computational cost. However, because its decisions are based solely on the current electricity price and fixed threshold values, it cannot proactively account for future PV generation, future EV charging demand, or distribution network power flow constraints. Compared with the no-battery rule-based controller, this strategy allows the impact of a simple price-responsive battery operation on the total operating cost to be evaluated. Compared with the MADRL and MPC approaches, it serves as a baseline that requires neither learning nor optimization.

3 Experimental Setup

This chapter describes the data, device parameters, training parameters, grid configuration, and evaluation metrics used in the experiments.

3.1 Dataset and Simulation Period

The data used in this chapter is located in the project directory `datasets/prosumer/`. The data includes household load, PV reference generation, and electricity price. The load profiles for the three prosumers are

SFH12, SFH18, SFH20.

Training and evaluation use different years and date ranges. The current configuration is shown in Table 3.1.

Table 3.1: Dataset and Simulation Time Period.

Item	Setting	Description
Data Directory	<code>datasets/prosumer</code>	Load, PV, and electricity price data
Training Year	2019	Used for LSTM and MADRL training
Training Dates	2019-01-01 to 2019-12-31	Main training data period
Evaluation Year	2020	Used for controller evaluation
Evaluation Dates	2020-04-01 to 2020-04-15	Main evaluation data period
Time Step	0.25 h	15-minute resolution
Daily Steps	96	One natural day episode
Load Profiles	SFH12, SFH18, SFH20	Three prosumers

During the evaluation phase, 15 evaluation episodes are selected by default. Each episode corresponds to the operation process of one day, where the controller reads observations and outputs actions at each time step, and the environment updates the system state.

3.2 Battery and EV Parameters

Battery parameters are shown in Table 3.2.

Table 3.2: Stationary battery parameters.

Parameter	Value	Description
C_i^b	100 kWh	Battery capacity for each prosumer
$SOC_{\min}^b$	0.05	Minimum battery SoC
$SOC_{\max}^b$	0.95	Maximum battery SoC
SOC_0^b	0.05	Initial battery SoC for evaluation
η_b	0.95	Battery charging/discharging efficiency
$r_{\max}^b$	0.5	Maximum charging/discharging rate
$P_i^{b,\max}$	50 kW	Maximum charging/discharging power

The maximum power of the battery is given by its capacity and maximum charging rate:

$$P_i^{b,\max} = r_{\max}^b C_i^b = 0.5 \times 100 = 50 \text{ kW}.$$

This setting allows the battery to charge when prices are low or PV generation is sufficient, and discharge when prices are high or load is significant, while keeping the SoC within the bounds $[0.05, 0.95]$.

EV parameters are shown in Table 3.3.

Table 3.3: EV parameters.

Parameter	Value	Description
C_i^{ev}	60 kWh	Battery capacity for each EV
$SOC_{\min}^{ev}$	0.10	Minimum EV SoC
$SOC_{\max}^{ev}$	0.95	Maximum EV SoC
SOC_{arr}^{ev}	0.30	SoC at home arrival
SOC_{req}^{ev}	0.90	Required SoC at departure
$P_i^{ev,\max}$	11 kW	Maximum charging power for home charging
η_{ev}	0.95	EV charging efficiency
t^{arr}	72	Home arrival time (18:00)
t^{dep}	28	Home departure time (07:00)

3.3 Training Parameters

3.3.1 LSTM Training Parameters

The LSTM forecasting module predicts the electricity price, load demand, and PV generation. The historical input window is set to

$$h = 192,$$

corresponding to the previous 48 hours, while the prediction horizon is

$$H = 49,$$

corresponding to the current time step and approximately the following 12 hours. The LSTM training parameters are summarized in Table 3.4.

Table 3.4: LSTM forecasting model training parameters.

Parameter	Value
history window	192
sequence length	49
batch size	1024
train ratio	0.70
validation ratio	0.15
price epochs	50
price hidden size	128
price layers	2
load epochs	20
load hidden size	64
PV epochs	20
PV hidden size	64

3.3.2 MADRL Training Parameters

The training parameters of the MADRL framework are summarized in Table 3.5.

The MADRL approaches compared in the experiments include `MADRL_BASE`, which does not incorporate any safety mechanism, and `MADRL_PROJECTION`, which applies a safety projection before action execution.

Table 3.5: MADRL training parameters.

Parameter	Value
action dimension	3
hidden dimension	256
discount factor γ	0.999
soft update τ	0.01
policy update frequency	2
policy noise	0.2
noise clip	0.5
train episodes	20
number of environments	1
replay buffer size	50000
batch size	128
actor learning rate	10^{-4}
critic learning rate	10^{-4}
n-step return	48
exploration noise	0.4 to 0.2

3.4 Network Configuration

The network security limits adopted in the experiments are summarized in Table 3.6.

3.5 Evaluation Metrics

Evaluation metrics can be divided into four categories: economic metrics, grid safety metrics, EV departure SoC metrics, and diagnostic metrics for the relationship between EV/battery and electricity prices. These metrics are used to evaluate the controller’s performance from different perspectives, but this chapter only defines the metrics without discussing specific results.

3.5.1 Economic Metrics

The economic metrics are used to evaluate the operating cost of the controller and the economic benefits achieved through energy storage. The main metrics include:

$$C_{total}^{bat} = \sum_{t,i} \max(P_{i,t}^b, 0) \tilde{p}_t \Delta t - \sum_{t,i} \max(-P_{i,t}^b, 0) \tilde{p}_t \Delta t,$$

Table 3.6: Network configuration and security limits.

Parameter	Value
SimBench code	1-LV-rural1--0-sw
agent bus ids	12, 4, 2
power flow solver	Newton-Raphson
$V_{\min}$	0.95 p.u.
$V_{\max}$	1.05 p.u.
line loading limit	100%
transformer loading limit	100%

$$C_{total}^{ev} = \sum_{t,i} P_{i,t}^{ev} \tilde{p}_t \Delta t,$$

$$C^{total} = C_{total}^{bat} + C_{total}^{ev} + C_{total}^{pv}.$$

Besides the total cost, the records also include storage charge cost, storage discharge revenue, storage profit, and episode reward.

3.5.2 Grid Safety Metrics

Grid safety metrics are derived from the pandapower power flow results. They mainly include:

$$\begin{aligned} & \min V_{b,t}, \quad \max V_{b,t}, \\ & \text{voltage violation steps,} \\ & \text{trafo loading max pct,} \quad \text{trafo overload steps,} \\ & \text{feeder netload ramp mean abs.} \end{aligned}$$

where voltage violation steps represents the number of time steps where at least one bus voltage violates the bounds; trafo loading max pct represents the maximum transformer loading during the evaluation; feeder netload ramp is used to measure the severity of changes in the feeder’s net load.

3.5.3 EV Departure SoC Metrics

EV metrics are used to evaluate whether users’ travel demands are met. The core metrics include departure SoC, departure gap, EV session energy, and EV session cost. The departure SoC gap is defined as

$$g_i^{dep} = \max(0, SOC_{req}^{ev} - SOC_{i,t^{dep}}^{ev}).$$

If $g_i^{dep} = 0$, it indicates that agent i meets the target SoC when departing; if $g_i^{dep} > 0$, it indicates that the departure demand is not fully satisfied. The evaluation also records the total charging amount for each EV session:

$$E_i^{ev,session} = \sum_{t \in \mathcal{T}^{ev}} P_{i,t}^{ev} \Delta t,$$

3.5.4 Relationship between EV/Battery and Electricity Prices

To analyze whether the controller has learned price-aware scheduling, this paper also records the time-aligned sequences of EV charging power, battery charging/discharging power, and import price.

$$P_{i,t}^{ev} \text{ vs. } \tilde{p}_t,$$

$$P_{i,t}^b \text{ vs. } \tilde{p}_t.$$

A controller with good price responsiveness should schedule EV charging and battery charging predominantly during periods of low electricity prices, while battery discharging should occur primarily during periods of high electricity prices.

4 Results and Discussion

This chapter presents and analyzes the experimental results. Before evaluating the performance of the proposed MADRL framework, the forecasting accuracy of the LSTM model for electricity prices is first analyzed, together with the potential impact of forecasting errors on the subsequent MADRL scheduling decisions. Next, the convergence behavior of the MADRL algorithm is investigated. Finally, the performance of MADRL under different control strategies is comprehensively evaluated and discussed.

4.1 LSTM Price Prediction Results

Figure 4.1 presents the LSTM electricity price forecasts used in the seven-day MADRL experiment. The blue curve represents the ground-truth electricity price under the perfect forecast setting, while the red dashed curve represents the electricity price predicted by the LSTM model. The lower subplot further illustrates the corresponding prediction error.

To evaluate the accuracy of LSTM price prediction, this thesis adopts MAE and RMSE as evaluation metrics. Let p_t represent the perfect electricity price at time step t , and $\hat{p}_t$ represent the electricity price predicted by the LSTM model. Let T be the number of evaluation time steps. The two metrics are defined as

$$MAE = \frac{1}{T} \sum_{t=1}^T |\hat{p}_t - p_t|,$$
$$RMSE = \sqrt{\frac{1}{T} \sum_{t=1}^T (\hat{p}_t - p_t)^2}.$$

The MAE of the LSTM electricity price forecast is 0.0174 EUR/kWh, while the RMSE is 0.0212 EUR/kWh. This indicates that the LSTM model is able to represent the overall variation trend of the electricity price with reasonable accuracy.

However, the curves show that certain deviations remain around local peaks and valleys. In high-price periods, the electricity prices predicted by the LSTM are generally lower than the corresponding peaks under the perfect forecast setting, while the difference between the maximum and minimum values is also reduced. In other words, the

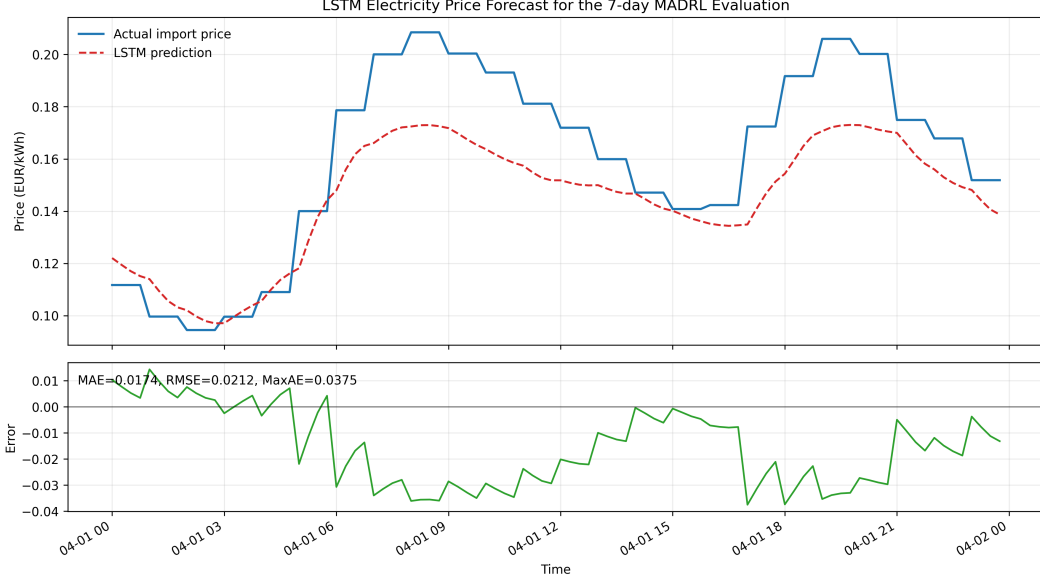


Figure 4.1: LSTM Price Prediction Results vs. Perfect Price

predicted electricity price exhibits a certain smoothing effect on the original price fluctuations. When high electricity prices are underestimated, the MADRL controller may underestimate the economic benefit of battery discharging or reducing electricity imports. When low electricity prices are overestimated, the controller may reduce battery charging or EV charging during low-price periods.

Overall, the LSTM forecast is able to capture the general trend of electricity price. Nevertheless, prediction errors around local peaks and valleys may affect the coordination among the energy storage system, EV, and PV generation, thereby influencing the MADRL analysis results.

4.2 MADRL Convergence Analysis

During the development process, the MADRL_BASE controller was trained for both 100 episodes and 1000 episodes to investigate the influence of the training duration on convergence. Figure 4.2 presents the corresponding training curves for these two training settings. The upper subplot shows the total reward obtained in each episode together with its moving-average curve, while the lower subplot illustrates the evolution of the critic loss. To better visualize the overall training trend, a moving average over 10 episodes is applied to the 100-episode experiment, whereas a moving average over 50 episodes is used for the 1000-episode experiment.

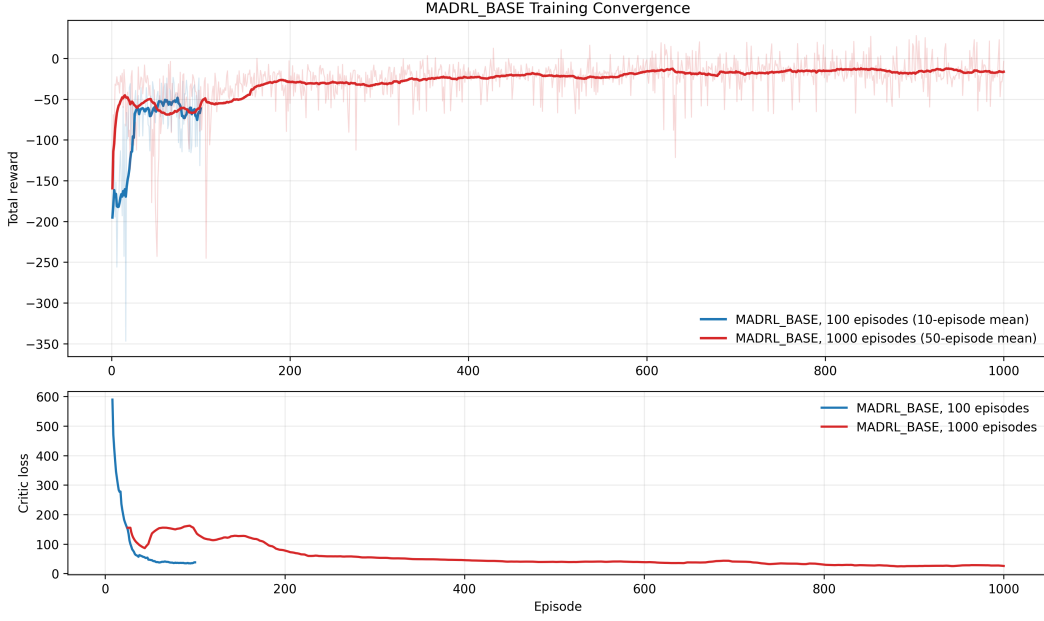


Figure 4.2: Convergence Comparison of MADRL_BASE under Different Numbers of Training Episodes

As shown in Figure 4.2, when the training duration is limited to only 100 episodes, the total reward still exhibits significant fluctuations throughout the training process. The total reward improves from -195.19 in the first episode to -50.43 in the 100th episode. Although the reward has improved considerably compared with the beginning of training, the average total reward over the last 10 episodes is still only -61.38 , with a standard deviation of 38.90. This indicates that the training process has not yet reached a stable convergence by the end of the 100-episode experiment.

In contrast, the 1000-episode experiment shows better convergence behavior. The total reward increases from -159.30 in the first episode to -3.57 in the 1000th episode. During the final stage of training, the average total reward over the last 50 episodes reaches -16.33 , with a standard deviation of 18.96. Compared with the 100-episode experiment, the average reward at the end of training is higher, while the fluctuations are smaller. Furthermore, the critic loss in the 1000-episode experiment remains relatively stable during the later stages of training. The critic loss of the final episode is 20.03, and the average critic loss over the last 50 episodes is 25.84. These results indicate that the critic gradually learns a stable estimate of the state-action value function, enabling the actor to develop a more consistent control policy.

Therefore, a training of 1000 episodes allows the MADRL to more effectively learn

the dynamic relationships among electricity prices, load demand, PV generation, battery SoC, and EV charging behavior. Consequently, all subsequent experiments adopt 1000 training episodes as the default MADRL training configuration.

4.3 EV soft constraint results

This section analyzes the performance of different MADRL controllers under EV soft constraint strategy. In the soft constraint approach, the coordination among the stationary battery, EV charging, and PV generation is achieved only through the design of the reward function. The analysis focuses on the influence of different reward components and safety correction mechanisms on the economic performance, EV departure SoC, electricity price responsiveness, and distribution network security.

The comparison follows the controller development process. First, MADRL_BASE is used as the initial benchmark. Next, the EV cost weight is introduced to increase the controller’s awareness of EV charging costs. Subsequently, an EV progress penalty is incorporated to encourage the EV to gradually approach the required departure SoC throughout the connection window. Finally, to ensure distribution network security, the safety projection mechanism is evaluated, together with an additional emergency charging strategy to investigate its impact on the soft-constrained controller. The corresponding baseline methods include ADMM-MPC Soft, Local MPC Soft, and the rule-based controllers.

First, Table 4.1 summarizes the economic performance of the different control methods.

Table 4.1: Economic Performance Comparison of Different EV Soft Constraint Controllers and Baseline Methods

Controller	Total cost	Battery storage profit	EV charging cost
MADRL Base	57.39	272.39	329.78
Base + Cost Weight	251.64	102.75	354.39
Base + Progress	188.69	146.98	335.68
Projection + Progress	462.56	-109.01	353.55
Projection + EV EME	272.61	32.66	305.28
Base + EV EME	94.28	100.93	195.22
ADMM-MPC Soft	-24.86	235.55	210.69
Local MPC Soft	-83.25	291.58	208.32
Rule-based + Battery	274.01	112.89	386.90
Rule-based No Battery	386.90	0.00	386.90

Figure 4.3 compares the total economic performance of the different control strategies. It can be observed that **MADRL_BASE** achieves a total cost of 57.39 EUR, which is the lowest among all MADRL soft constraint controllers and also outperforms the rule-based controller. This superior economic performance is mainly attributed to its strong battery arbitrage capability, resulting in a storage profit of 272.39 EUR. However, the low total cost of **MADRL_BASE** does not necessarily indicate the best overall control performance. As shown in Table 4.2, the average departure SoC is 0.9460, while the minimum departure SoC is only 0.8782. Furthermore, the departure SoC falls below the required target on 2 occasions, resulting in a cumulative departure SoC gap of 0.0376. In other words, although **MADRL_BASE** achieves favorable economic performance through battery arbitrage, it does not fully satisfy the EV departure SoC requirement.

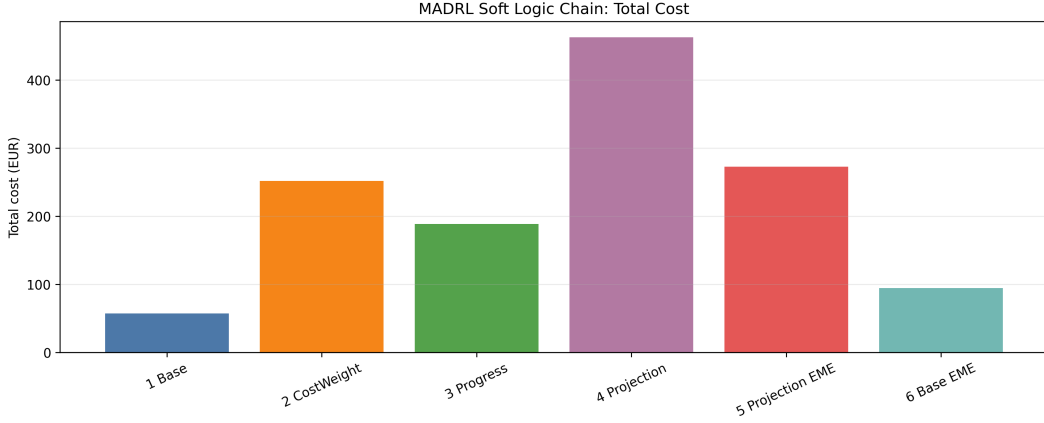


Figure 4.3: Comparison of the Total Cost of Different EV Soft Constraint Controllers

To increase the importance of the EV charging demand in the reward function, the Cost Weight strategy is first introduced based on **MADRL_BASE**, where the EV cost weight is set to 1.5. Experimental results show that this strategy is able to fully satisfy the EV departure SoC requirement. However, the total cost increases from 57.39 EUR for **MADRL_BASE** to 251.64 EUR. Meanwhile, the storage profit decreases from 272.39 EUR to 102.75 EUR, and the actual EV charging cost increases from 329.78 EUR to 354.39 EUR. The reason is that simply increasing the EV cost weight changes the relative importance of EV charging behavior with respect to battery arbitrage in the reward function, but it does not explicitly guide the agent on how to gradually satisfy the departure SoC requirement during the EV connection window. To avoid departure-related penalties, the agent tends to complete EV charging earlier and adopts a more conservative charging strategy. At the same time, the battery spends more effort coordinating with the grid conditions rather than performing independent price arbitrage, resulting in a significant

Table 4.2: Comparison of EV Departure SoC and Feasibility Metrics of Different EV Soft Constraint Controllers

Controller	Mean dep. SoC	Below target	Dep. gap
MADRL Base	0.9460	2	0.0376
Base + Cost Weight	0.9500	0	0.0000
Base + Progress	0.9500	0	0.0000
Projection + Progress	0.9463	1	0.0353
Projection + EV EME	0.8702	16	2.7520
Base + EV EME	0.6112	45	12.9948
ADMM-MPC Soft	0.8961	45	0.1745
Local MPC Soft	0.8963	45	0.1669
Rule-based + Battery	0.9500	0	0.0000
Rule-based No Battery	0.9500	0	0.0000

reduction in storage profit. In other words, the Cost Weight strategy successfully resolves the EV departure SoC issue and provides better operational security, but at the expense of the economic performance of the battery.

The progress penalty weight is then introduced, corresponding to the **Base + Progress** experiment. Unlike the Cost Weight strategy, the progress term does not simply increase the reward associated with the EV charging cost. Instead, it provides a progressive constraint that encourages the EV SoC to gradually approach the required departure SoC throughout the connection window. As shown in Table 4.2, **Base + Progress** also achieves 0 violations of the departure SoC requirement, while both the average and minimum departure SoC reach 0.9500. The total cost is 188.69 EUR, which is lower than the 251.64 EUR obtained by **Base + Cost Weight**. In addition, the storage profit increases to 146.98 EUR, compared with 102.75 EUR for the Cost Weight strategy. Although the EV charging cost of **Base + Progress** is 335.68 EUR, which is slightly higher than the 329.78 EUR of **MADRL_BASE**, it is lower than that of the Cost Weight strategy. This improvement is achieved because the progress reward provides a clearer temporal guidance for EV charging, allowing the agent to satisfy the departure SoC requirement while preserving part of the battery arbitrage capability. Therefore, the **Base + Progress** strategy achieves a more balanced trade-off between EV feasibility and economic performance.

Figure 4.4 illustrates the EV departure SoC over the 15-day evaluation period for the different control strategies. The black dashed line represents the target departure SoC of 0.90. It can be observed that **Base + Progress** remains above the target SoC throughout the entire evaluation period, whereas **MADRL_BASE** falls below the target on

several days. This further demonstrates that the progress term improves the ability of the soft-constrained controller to consistently satisfy the EV departure SoC requirement.

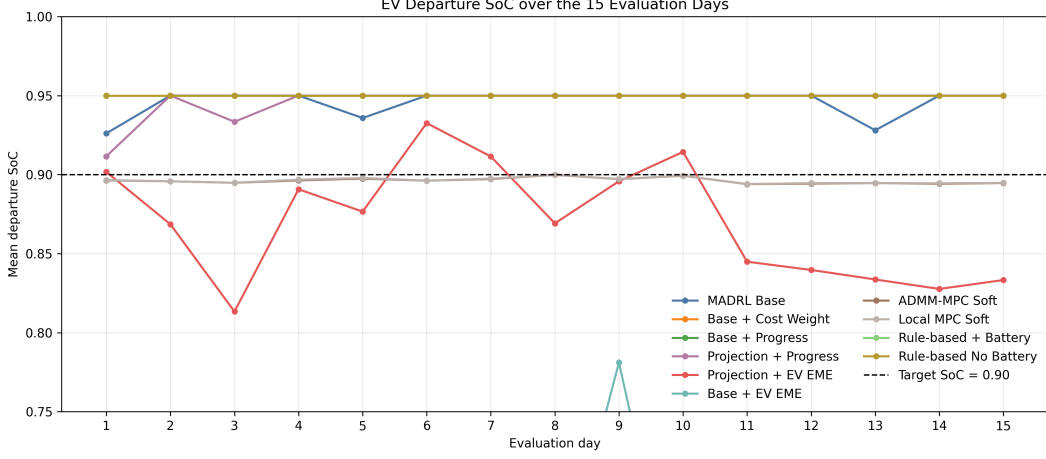


Figure 4.4: Comparison of EV Departure SoC over a 15-day Evaluation Period of EV soft Constraint Controllers

Table 4.3 summarizes the distribution network safety indicators of the different control strategies. **MADRL_BASE** exhibits 2 voltage violations, while the maximum transformer loading reaches 195.22%. This indicates that although **MADRL_BASE** achieves good economic performance, its aggressive battery charging and discharging actions may impose significant stress on the distribution network. Nevertheless, its distribution network security performance is still better than that of the rule-based controller and the Local MPC Soft controller. For **Base + Progress**, the number of voltage violations is reduced to 1, and the maximum transformer loading decreases to 136.97%, representing a clear improvement over **MADRL_BASE**. This suggests that the progress term not only improves EV feasibility but also indirectly mitigates extreme control actions. The main reason is that EV charging no longer relies heavily on compensatory charging near the departure time.

To further improve the distribution network security, the safety projection mechanism is incorporated, corresponding to the **Projection + Progress** experiment. After introducing the projection mechanism, the number of voltage violations is reduced to 0, while the maximum transformer loading decreases to 105.22%, resulting in better network security than both **MADRL_BASE** and **Base + Progress**. However, the total cost increases to 462.56 EUR, and the storage profit becomes -109.01 EUR, indicating that the battery not only fails to generate arbitrage profit but instead incurs a net cost. The reason is that the safety projection modifies the battery power and PV curtailment be-

Table 4.3: Comparison of Distribution Network Safety Indicators under the EV Soft Constraint Strategy

Controller	Voltage count	Min V	Max V	Max trafo
MADRL Base	2	0.9681	1.0508	195.22
Base + Cost Weight	0	0.9708	1.0477	157.42
Base + Progress	1	0.9675	1.0527	136.97
Projection + Progress	0	0.9746	1.0451	105.22
Projection + EV EME	0	0.9710	1.0462	103.81
Base + EV EME	1	0.9705	1.0501	206.11
ADMM-MPC Soft	0	0.9736	1.0486	97.75
Local MPC Soft	19	0.9663	1.0510	168.48
Rule-based + Battery	3	0.9682	1.0511	128.76
Rule-based No Battery	0	0.9707	1.0306	125.63

fore the actions are executed to satisfy the voltage and transformer loading constraints. For the MADRL actor, the projection changes the relationship between its original actions and the resulting environmental feedback, causing the battery arbitrage strategy to be frequently clipped or corrected during execution. In particular, when the distribution network operates close to its security limits, battery actions originally intended for charging during low-price periods and discharging during high-price periods may be restricted by the projection mechanism, preventing the battery from fully responding to electricity price signals. Therefore, although **Projection + Progress** significantly improves distribution network security, it sacrifices both the battery learning capability and the economic performance.

Figure 4.5 compares the number of voltage violations and the maximum transformer loading for the different control strategies. In addition, Figure 4.6 illustrates the relationship between the battery power and the electricity price on a representative day for the different MADRL soft constraint controllers. Compared with **MADRL_BASE** and **Base + Progress**, the battery power profile of **Projection + Progress** is subject to much stronger restrictions.

Table 4.4 further summarizes the relationship between EV charging and the electricity price intervals. The weighted EV charging prices of ADMM-MPC Soft and Local MPC Soft are 0.1244 EUR/kWh and 0.1229 EUR/kWh, respectively, which are significantly lower than those of the MADRL soft constraint controllers. This indicates that, with a known optimization model and forecast information, the MPC-based controllers are better able to schedule EV charging during low-price periods. For **MADRL_BASE**, the weighted EV charging price is 0.1714 EUR/kWh, while 45.99% of the EV charging energy is sup-

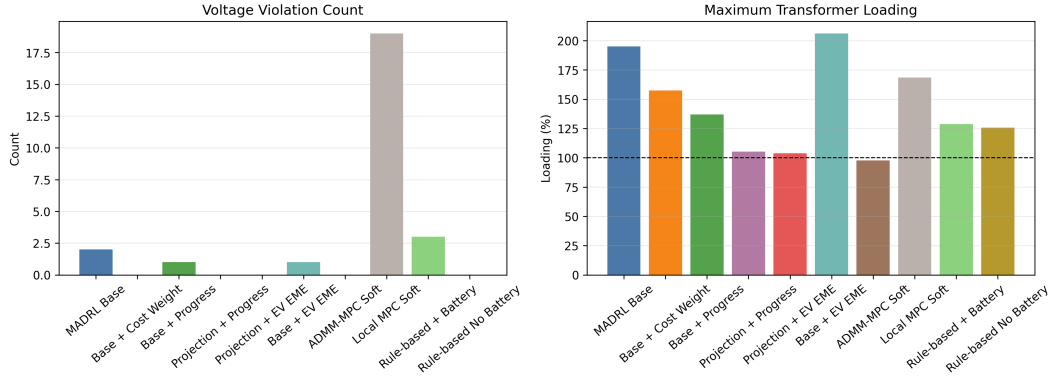


Figure 4.5: Comparison of Distribution Network Safety Indicators under the EV Soft Constraint Strategy

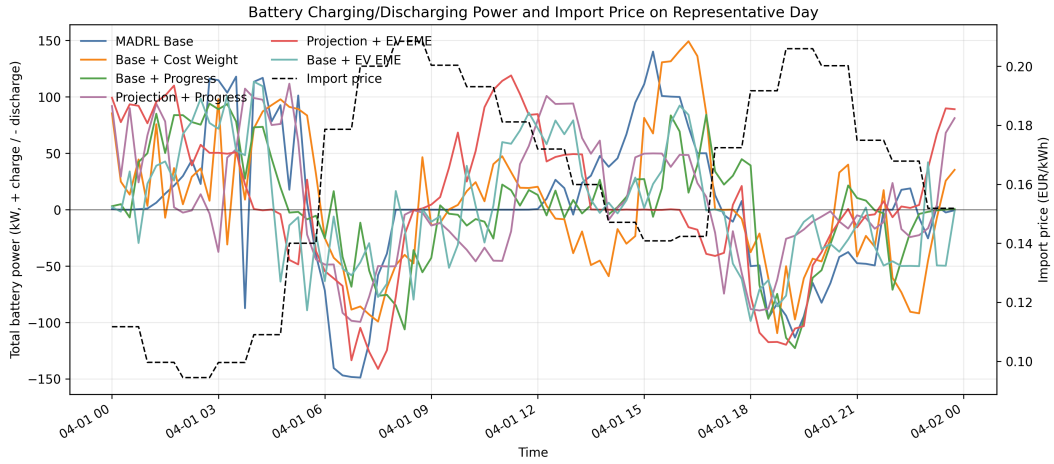


Figure 4.6: Relationship between Battery Power and Electricity Price under Different EV Soft Constraint Methods

plied during high-price periods. This suggests that its primary optimization objective remains battery arbitrage rather than low-cost EV charging. For **Base + Cost Weight**, the proportion of charging during high-price periods further increases to 50.82%, which explains why the actual EV charging cost increases despite the introduction of the EV cost weight. The weighted EV charging price of **Base + Progress** is 0.1723 EUR/kWh, with 45.75% of the charging energy occurring during high-price periods, which is close to that of **MADRL_BASE**. However, its main advantage lies in consistently satisfying the EV departure SoC requirement while simultaneously improving the distribution network security.

Table 4.4: Relationship between EV Charging and Electricity Price Intervals under Different EV Soft Constraint Methods

Controller	Weighted price	EV energy	Low share	High share
MADRL Base	0.1714	1924.17	0.2761	0.4599
Base + Cost Weight	0.1808	1960.14	0.2698	0.5082
Base + Progress	0.1723	1947.67	0.2960	0.4575
Projection + Progress	0.1819	1943.97	0.2687	0.5153
Projection + EV EME	0.1466	2083.08	0.3380	0.3069
Base + EV EME	0.1487	1312.62	0.3383	0.3218
ADMM-MPC Soft	0.1244	1694.25	0.4529	0.2278
Local MPC Soft	0.1229	1694.72	0.4679	0.2236
Rule-based + Battery	0.1963	1970.53	0.2333	0.6005
Rule-based No Battery	0.1963	1970.53	0.2333	0.6005

In addition, Figure 4.7 illustrates the relationship between the EV charging power and the electricity price on a representative day. It can be observed that, under the soft constraint strategy, EV charging is not always scheduled during the lowest-price periods. Instead, the controller must simultaneously consider the battery SoC, PV generation, load demand, and the EV-related constraint terms in the reward function. Therefore, compared with the MPC-based controllers, the MADRL soft constraint controllers still have room for improvement in scheduling EV charging during low-price periods.

This section also evaluates emergency charging, which is introduced to ensure that the EV departure SoC can meet the required level. The total cost of **Projection + EV EME** is 272.61 EUR, with a storage profit of 32.66 EUR, while the total cost of **Base + EV EME** is 94.28 EUR, with a storage profit of 100.93 EUR. From an economic perspective, the total cost of **Base + EV EME** is lower than that of **Base + Progress**. However, its EV departure SoC performance is significantly worse: the average departure SoC is only 0.6112, the minimum departure SoC is 0.4985, the number of departures below the target

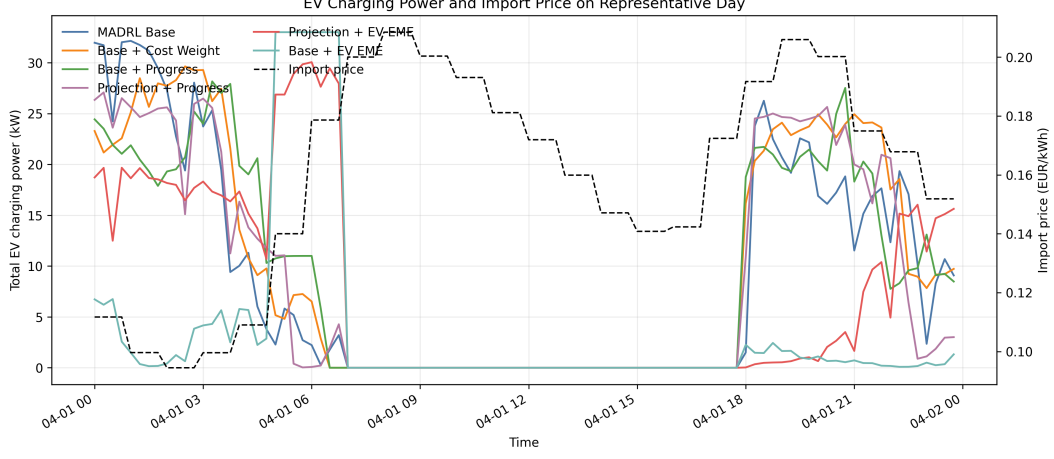


Figure 4.7: Relationship between EV Charging Power and Electricity Price under Different EV Soft Constraint Methods

SoC reaches 45, and the cumulative departure gap reaches 12.9948. For **Projection + EV EME**, the average departure SoC is 0.8702, the minimum departure SoC is 0.5400, and the number of departures below the target SoC is 16. These results indicate that emergency charging does not consistently resolve the EV departure requirement under the current soft constraint setting.

The reason for this behavior is that emergency charging is a remedial mechanism rather than a replacement for the main learning signal provided by the soft reward. In the soft constraint controller, if the actor has not learned to continuously satisfy the EV progress requirement during the connection window, emergency charging may only supplement energy at local time steps and cannot establish a stable long-term scheduling pattern. At the same time, the additional charging introduced by EME reduces the available space for battery arbitrage and may alter the local network power distribution. The maximum transformer loading of **Base + EV EME** reaches 206.11%, which is even higher than the 195.22% of **MADRL_BASE**, indicating that its safety performance is not satisfactory. Therefore, the soft constraint strategy combined with EME is not suitable as the main MADRL soft constraint controller in this thesis.

Finally, Figure 4.8 uses **Base + Progress** as an example to illustrate the EV charging power and battery charging and discharging power of each agent. The figure shows that, when **Base + Progress** is selected as the main soft constraint controller, a coordination relationship still exists between EV charging and battery actions within each prosumer. The EV must gradually approach the target SoC during the connection window, while the battery adjusts its charging and discharging behavior according to the electricity

price and local load. This coordination is the main advantage of the MADRL soft constraint controller over the simple rule-based method.

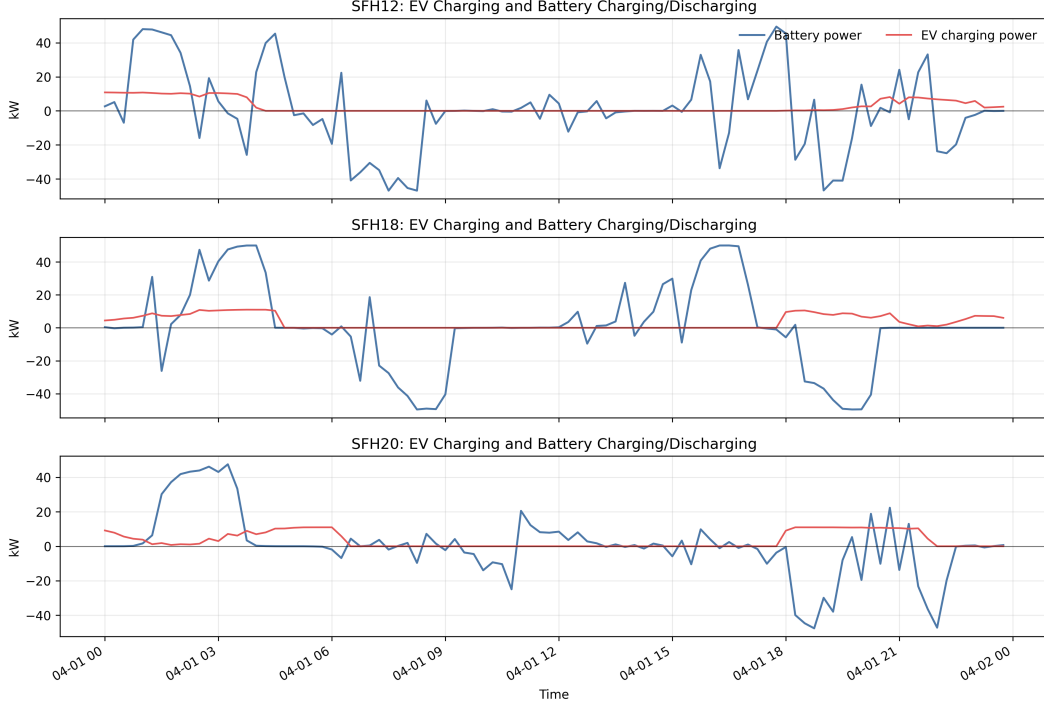


Figure 4.8: Base + Progress 中每个 agent 的 EV 充电与电池充放电功率

Overall, although MADRL_BASE achieves a relatively low total cost through strong battery arbitrage, it exhibits deficiencies in both the EV departure SoC and distribution network security. Although Base + Cost Weight satisfies the EV departure requirement, its excessive emphasis on EV-related costs weakens the battery arbitrage capability, resulting in a higher total cost. Projection + Progress achieves the best distribution network security performance; however, the projection mechanism imposes overly strong modifications on the battery actions, causing the storage profit to become negative. Under the soft constraint setting, the EME strategy cannot consistently guarantee the EV departure SoC and may even deteriorate the distribution network security. In comparison, Base + Progress simultaneously satisfies the EV departure SoC requirement, improves the distribution network security of MADRL_BASE, and maintains a certain level of battery profitability.

4.4 EV hard constraint results

This section further analyzes the experimental results of the EV hard constraint controllers. Unlike the soft constraint controllers, the hard constraint controllers treat the EV departure SoC requirement as a hard constraint during action execution. As described in Chapter 4, the target EV departure SoC is guaranteed through hard projection or feasibility correction within the EV connection window. Therefore, the main challenge of the hard constraint controllers is no longer whether the EV departure requirement can be learned through the reward function, but rather how to balance battery arbitrage, electricity price responsiveness, and distribution network security while satisfying the EV hard constraint.

This section analyzes the experimental results following the order of the controller development process. First, the basic **MADRL_BASE** controller is evaluated. Then, to improve distribution network security and increase the attention paid to the EV charging cost, the distribution network projection and the EV cost weight are introduced. Subsequently, the EV price-aware penalty is further incorporated, resulting in the **Projection + PriceAware** controller. Finally, **Projection Hard** is compared with **Projection Hard** under the setting of $\text{SOC}_{\max}^{\text{ev}} = 0.91$. Table 4.5 summarizes the corresponding economic performance indicators.

Table 4.5: Economic Performance Comparison of EV Hard Constraint Controllers and Baseline Methods

Controller	Total cost	EV cost	Storage profit	Weight
Base hard	81.67	287.53	205.87	1.0
Projection + CostWeight	356.28	289.48	-66.81	1.5
Projection + PriceAware	238.89	291.35	52.46	1.0
Projection Hard	186.04	373.42	187.39	1.0
Projection Hard (SOCmax=0.91)	494.25	288.43	-205.82	1.0
ADMM-MPC Hard	-2.86	235.74	238.60	1.0
Local MPC Hard	-36.37	235.68	272.06	1.0
Rule-based + Battery	274.01	386.90	112.89	1.0
Rule-based No Battery	386.90	386.90	0.00	1.0

As shown in Table 4.5, **Base Hard** achieves the best overall economic performance among the EV hard constraint controllers, with a total cost of 81.67 EUR, a storage profit of 205.87 EUR, and an EV charging cost of 287.53 EUR. Compared with **Base + Progress** in the EV soft constraint controllers, the hard constraint controller can satisfy the EV departure requirement more directly while still retaining strong battery

arbitrage capability. Under **Base Hard**, the battery charging cost is 908.19 EUR, and the battery discharging revenue reaches 1114.06 EUR. Moreover, under the **Base Hard** strategy, both the total electricity cost and the EV charging cost are lower than those of the **Rule-based + Battery** controller, whose EV charging cost is 386.90 EUR.

However, the main limitation of **Base Hard** is that its distribution network security remains insufficient. As shown in Table 4.6, this controller exhibits 1 voltage violation, a maximum transformer loading of 156.26%, and 19 transformer overload steps. This indicates that the EV hard constraint itself only guarantees the EV departure SoC and does not automatically ensure distribution network security. When the battery performs high-power charging and discharging for electricity price arbitrage, voltage violations or transformer overloads may still occur.

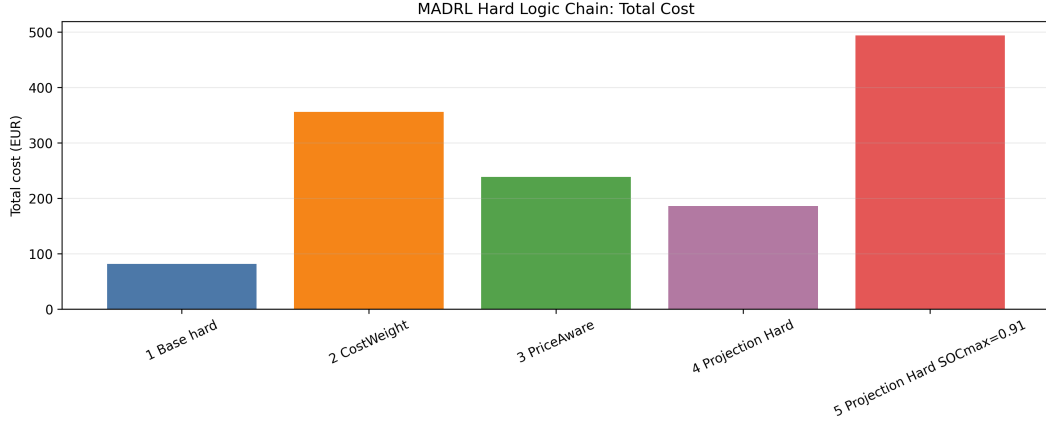


Figure 4.9: EV hard constraint 控制器设计逻辑链下的 total cost 对比

To simultaneously improve distribution network security and reduce the EV charging cost, the distribution network safety projection and the EV cost weight are introduced, resulting in the **Projection + CostWeight** controller. This controller reduces the number of voltage violations to 0, decreases the maximum transformer loading from 156.26% under **Base Hard** to 121.10%, and reduces the number of transformer overload steps from 19 to 1. Therefore, from the perspective of distribution network security, the projection mechanism is indeed effective. However, the total cost increases to 356.28 EUR, while the storage profit becomes -66.81 EUR. The fundamental reason is that the projection mechanism and the cost weight jointly reduce the available space for battery arbitrage. The battery charging cost is 759.31 EUR, whereas the battery discharging revenue is only 692.50 EUR, resulting in a negative storage profit. Although the EV charging cost is 289.48 EUR, which is close to the 287.53 EUR of **Base Hard**, the storage profit decreases from 205.87 EUR to -66.81 EUR, leading to a substantial deterioration in the

Table 4.6: Grid Safety Comparison of EV Hard Constraint Controllers and Baseline Methods

Controller	Voltage count	Min V	Max V	Max trafo
Base hard	1	0.9718	1.0505	156.26
Projection + CostWeight	0	0.9750	1.0447	121.10
Projection + PriceAware	0	0.9722	1.0460	103.56
Projection Hard	0	0.9707	1.0456	108.54
Projection Hard (SOCmax=0.91)	0	0.9722	1.0461	101.62
ADMM-MPC Hard	0	0.9736	1.0486	118.94
Local MPC Hard	25	0.9663	1.0549	163.63
Rule-based + Battery	3	0.9682	1.0511	128.76
Rule-based No Battery	0	0.9707	1.0306	125.63

overall economic performance.

Subsequently, the cost weight strategy is abandoned and replaced by an EV price-aware penalty, with the objective of encouraging EV charging during low-price periods. This controller further improves the distribution network security, with the maximum transformer loading reduced to 103.56%, which is lower than that of the previous controllers. However, from an economic perspective, the total cost of **Projection + PriceAware** remains 238.89 EUR, which is still higher than that of **Base Hard**. Its EV charging cost is 291.35 EUR, which is also higher than the 287.53 EUR of **Base Hard**. Table 4.7 shows that the weighted EV charging price of **Projection + PriceAware** is 0.1614 EUR/kWh, which is slightly higher than the 0.1600 EUR/kWh of **Base Hard**. The proportion of EV charging during high-price periods decreases to 38.58%, compared with 40.34% under **Base Hard**. However, the proportion of charging during low-price periods also decreases from 33.05% under **Base Hard** to 30.89%. These results indicate that the price-aware penalty indeed reduces part of the charging during high-price periods, but it does not significantly increase the proportion of charging during low-price periods. Its effectiveness is jointly constrained by the EV hard projection, the EV connection window, and the distribution network safety correction.

In addition, the battery profitability of **Projection + PriceAware** remains limited. Its battery charging cost is 1024.16 EUR, while the battery discharging revenue is 1076.62 EUR, resulting in a storage profit of only 52.46 EUR, which is significantly lower than the 205.87 EUR achieved by **Base Hard**. In other words, the price-aware penalty mainly influences the EV charging behavior, whereas the battery actions remain constrained by the safety projection. Although the proportion of EV charging during high-price periods is reduced, the battery fails to recover the arbitrage capability observed under

Table 4.7: Relationship Between EV Charging and Electricity Price Intervals for EV Hard Constraint Controllers

Controller	Weighted price	EV energy	Low share	High share
Base hard	0.1600	1797.35	0.3305	0.4034
Projection + CostWeight	0.1604	1804.74	0.3404	0.3978
Projection + PriceAware	0.1614	1805.37	0.3089	0.3858
Projection Hard	0.1897	1968.33	0.2469	0.5693
Projection Hard (SOC _{max} =0.91)	0.1616	1784.41	0.2895	0.3767
ADMM-MPC Hard	0.1382	1705.27	0.3893	0.2688
Local MPC Hard	0.1382	1705.26	0.3893	0.2688
Rule-based + Battery	0.1963	1970.53	0.2333	0.6005
Rule-based No Battery	0.1963	1970.53	0.2333	0.6005

Base Hard. Consequently, the overall economic performance remains unsatisfactory.

Therefore, this thesis further evaluates a **Projection Hard** variant without introducing either the EV cost weight or the EV price-aware penalty. **Projection Hard** achieves a total cost of 186.04 EUR and a storage profit of 187.39 EUR, which is close to the 205.87 EUR of **Base Hard**. At the same time, it reduces the number of voltage violations to 0, while the maximum transformer loading is limited to 108.54%. These results indicate that, for the hard constraint controller, directly applying the projection mechanism to enforce the distribution network safety constraints is more effective and stable than additionally introducing the EV cost weight or the EV price-aware penalty. The battery charging cost under this controller is 985.61 EUR, while the battery discharging revenue reaches 1172.99 EUR, indicating that the controller still preserves a strong capability for charging during low-price periods and discharging during high-price periods.

However, the EV charging cost of **Projection Hard** is relatively high, reaching 373.42 EUR. As shown in Tables 4.8 and 4.7, the average departure SoC and the minimum departure SoC are both 0.9500. This indicates that, under this hard projection setting, the EV tends to be charged close to the upper SoC limit of 0.95. Although this fully satisfies the departure requirement, it also results in unnecessary additional charging energy and a higher proportion of charging during high-price periods. Consequently, its EV charging cost is significantly higher than those of **Base Hard** and **Projection + PriceAware**.

To reduce the tendency of EV overcharging, this thesis further evaluates **Projection Hard** with $\text{SOC}_{\max}^{\text{ev}} = 0.91$. Under this controller, the average departure SoC is 0.9093, while the minimum departure SoC is 0.9000, which is much closer to the target departure SoC of 0.90 than the 0.9500 obtained with **Projection Hard**. At the same time, the EV

Table 4.8: EV Departure SoC and Feasibility Indices for EV Hard Controllers

Controller	Mean dep. SoC	Min dep. SoC	Below target	Projection gap
Base hard	0.9098	0.9000	1	9.72
Projection + CostWeight	0.9084	0.9000	6	71.25
Projection + PriceAware	0.9093	0.9000	3	51.13
Projection Hard	0.9500	0.9500	0	0.00
Projection Hard (SOC _{max} =0.91)	0.9093	0.9000	3	13.45
ADMM-MPC Hard	0.9000	0.9000	45	1680.31
Local MPC Hard	0.9000	0.9000	45	1672.26
Rule-based + Battery	0.9500	0.9500	0	0.00
Rule-based No Battery	0.9500	0.9500	0	0.00

charging cost decreases to 288.43 EUR, and the proportion of charging during high-price periods is reduced to 37.67%. From the perspective of EV charging, **Projection Hard** with $\text{SOC}_{\max}^{\text{ev}} = 0.91$ indeed reduces both overcharging and charging during high-price periods. However, its total cost increases to 494.25 EUR, while the storage profit becomes -205.82 EUR, representing the worst economic performance among all MADRL hard constraint controllers. The battery cost breakdown shows that the battery charging cost reaches 1172.91 EUR, whereas the battery discharging revenue is only 967.08 EUR. Although the battery still exhibits a relatively large average power magnitude of 40.79 kW, the charging and discharging timing no longer matches the electricity price, resulting in negative arbitrage.

This phenomenon indicates that, under the hard projection framework, simply reducing EV overcharging does not necessarily improve the overall economic performance. After the EV charging energy is reduced, the local net load and the battery SoC trajectory change accordingly, causing the projection mechanism to modify the battery actions differently. If the battery policy learned by the actor still depends on the original EV load profile, then under the new EV SoC upper bound or operation near the target SoC, the battery may charge during inappropriate electricity price periods or fail to discharge sufficiently during high-price periods, ultimately causing the storage profit to become negative. Therefore, although **Projection Hard** with $\text{SOC}_{\max}^{\text{ev}} = 0.91$ reduces the EV charging cost, it disrupts the battery arbitrage structure.

Based on the above results, **Base Hard** achieves the best economic performance among the MADRL hard constraint controllers, but it still suffers from distribution network security issues. **Projection + CostWeight** significantly improves the distribution network security, but the battery arbitrage becomes unprofitable. **Projection + PriceAware** reduces the proportion of charging during high-price periods to some extent, but it nei-

4 Results and Discussion

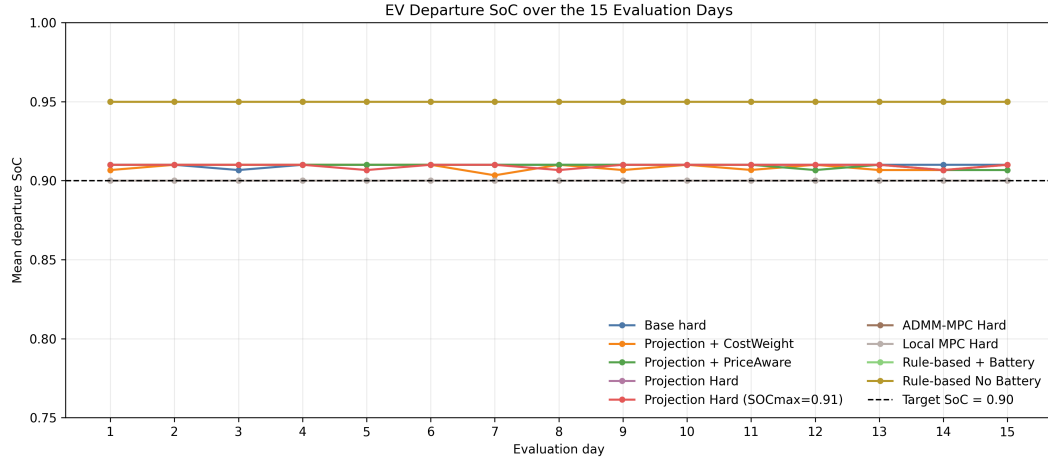


Figure 4.10: Comparison of EV Departure SoC among EV Hard Constraint Controllers over the 15-Day Evaluation Period

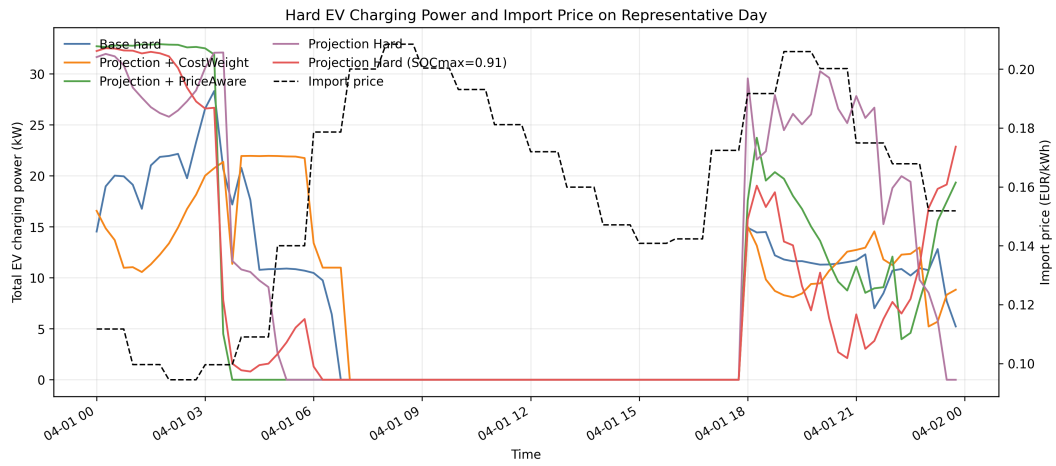


Figure 4.11: Comparison of EV Charging Power and Electricity Price among EV Hard Constraint Controllers over the 15-Day Evaluation Period

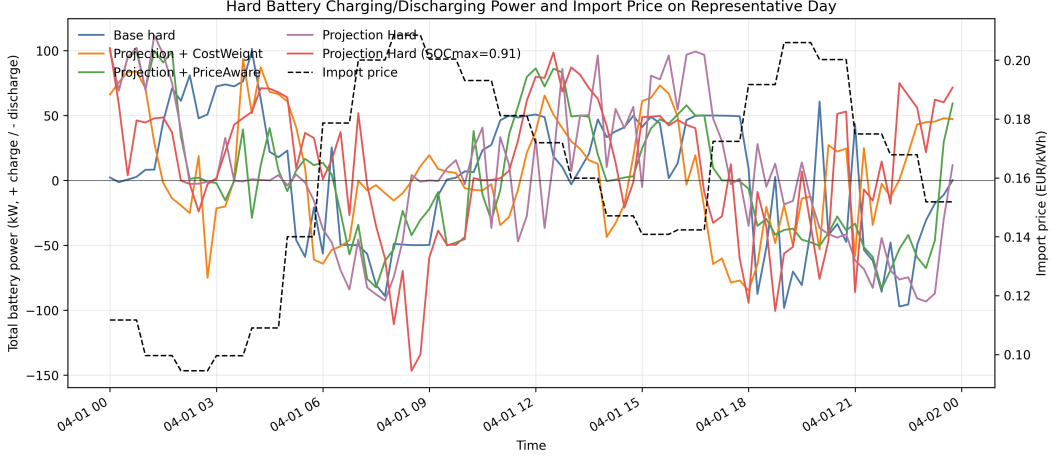


Figure 4.12: Comparison of Battery Charging/Discharging Power and Electricity Price among EV Hard Constraint Controllers over the 15-Day Evaluation Period

ther significantly reduces the EV charging cost nor restores the battery profitability. **Projection Hard** provides a good trade-off between distribution network security and battery profitability, but it exhibits EV overcharging and a relatively high EV charging cost. Although **Projection Hard** with $\text{SOC}_{\max}^{\text{ev}} = 0.91$ reduces the EV charging cost, it severely degrades the battery arbitrage capability. Therefore, under the hard constraint scenario, the controller design requires a trade-off among the EV hard constraint, distribution network security, battery arbitrage, and the EV charging cost. If economic performance is the primary objective, **Base Hard** is the preferred controller. If distribution network security and operational feasibility are prioritized, **Projection Hard** is a more robust choice.

4.5 MADRL Results

Based on the analyses presented in the previous two sections, this section further compares four representative controllers: **Soft Base** and **Soft Base + Progress** under the soft constraint scenario, together with **Base Hard** and **Projection Hard** under the hard constraint scenario. Among them, **Soft Base** and **Base Hard** represent the MADRL controllers with superior economic performance under their respective constraint frameworks. **Soft Base + Progress** represents the controller that improves the feasibility of the EV departure SoC under the soft constraint framework, while **Projection Hard** represents the controller that simultaneously emphasizes EV feasibility and distribution network security under the hard constraint framework. To further illustrate the

scheduling characteristics of these controllers, they are comprehensively compared with ADMM-MPC, Local MPC, **Rule-based + Battery**, and **Rule-based No Battery**.

Since the total cost has already been discussed separately in Section 4.3 and Section 4.4, it is not repeated here. Figure 4.13 illustrates the relationship between the EV charging power and the electricity price for the selected MADRL controllers over the 15-day evaluation period, while Figure 4.14 presents the relationship between the battery charging/discharging power and the electricity price. Table 4.9 further summarizes the average EV charging price, the average battery charging price, the average battery discharging price, and the price difference between battery charging and discharging.

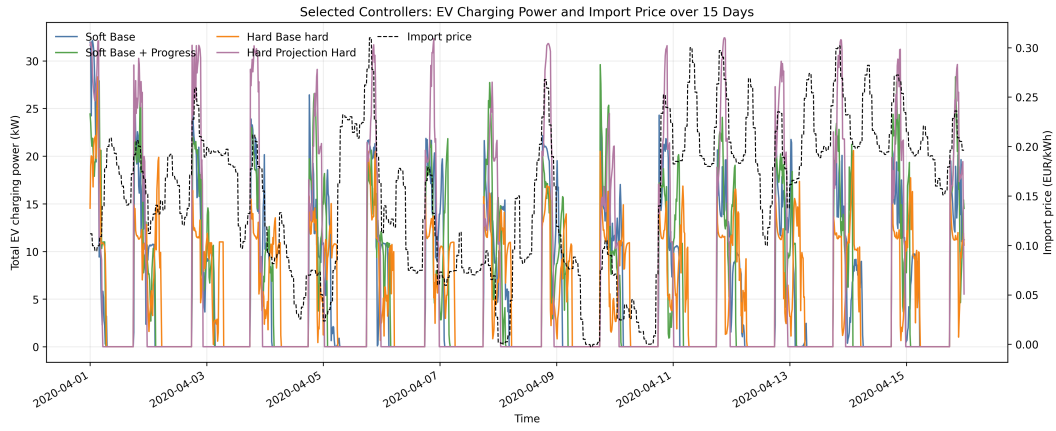


Figure 4.13: Comparison of EV Charging Power and Electricity Price among EV Hard Constraint Controllers over the 15-Day Evaluation Period

Table 4.9: Average EV and Battery Charging/Discharging Prices among EV Hard Constraint Controllers over the 15-Day Evaluation Period

Controller	EV price	EV energy	Battery charge	Battery discharge	Spread
Soft Base	0.1714	1924.17	0.1039	0.1589	0.0549
Soft Base + Progress	0.1723	1947.67	0.1300	0.1634	0.0334
Base hard	0.1600	1797.35	0.1275	0.1710	0.0435
Projection Hard	0.1897	1968.33	0.1295	0.1742	0.0447

As shown in Table 4.9, all four selected MADRL controllers are able to perform scheduling according to the electricity price signals to a certain extent. **Soft Base** achieves an average battery charging price of 0.1039 EUR/kWh and an average battery discharging price of 0.1589 EUR/kWh, resulting in a price difference of 0.0549 EUR/kWh, which is

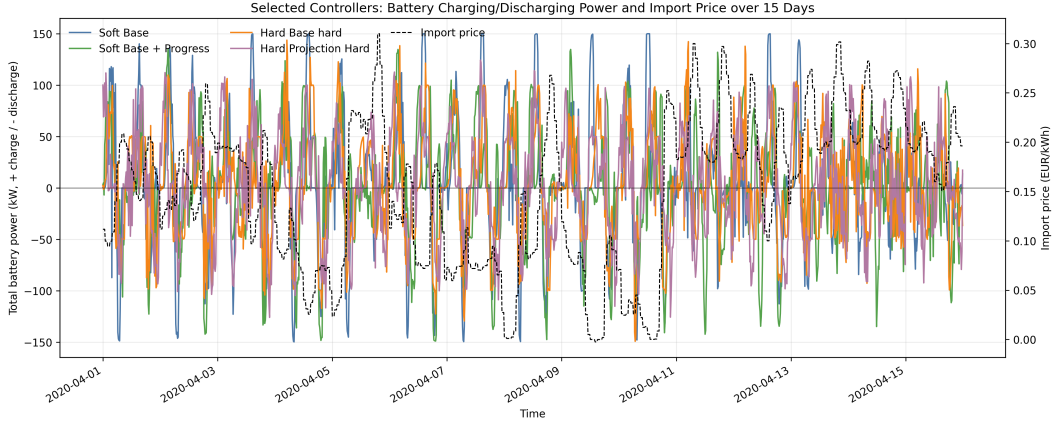


Figure 4.14: Comparison of Battery Charging/Discharging Power and Electricity Price among EV Hard Constraint Controllers over the 15-Day Evaluation Period

the largest among the four controllers. This indicates that **Soft Base** learns the battery arbitrage strategy most effectively, which also explains its relatively high storage profit. However, this controller mainly pursues economic benefits, while its EV departure SoC and distribution network security remain insufficient. Therefore, it cannot be regarded as a robust final controller by itself. The average EV charging price of **Soft Base + Progress** is 0.1723 EUR/kWh, which is close to the 0.1714 EUR/kWh of **Soft Base**. However, its battery charging–discharging price difference decreases to 0.0334 EUR/kWh, indicating that the progress penalty improves the EV charging progress but also reduces the battery’s flexibility to charge during low-price periods and discharge during high-price periods.

Base Hard achieves the lowest average EV charging price among the four selected MADRL controllers, at 0.1600 EUR/kWh, while maintaining a battery charging–discharging price difference of 0.0435 EUR/kWh. This indicates that, after incorporating the EV departure requirement into the action feasibility constraints, the hard constraint framework reduces part of the unnecessary EV charging during high-price periods while preserving a relatively strong battery arbitrage capability. In contrast, the average EV charging price of **Projection Hard** increases to 0.1897 EUR/kWh, while the total EV charging energy also reaches 1968.33 kWh. This is consistent with the analysis in Section 4.4: to ensure both EV feasibility and distribution network security, the controller tends to charge the EV to a higher SoC, thereby increasing the EV charging cost. Nevertheless, the average battery charging price and average battery discharging price of **Projection Hard** are 0.1295 EUR/kWh and 0.1742 EUR/kWh, respectively, resulting in a price difference of 0.0447 EUR/kWh. This indicates that the controller still preserves a certain

level of battery arbitrage capability under the projection constraints.

Compared with **Rule-based + Battery**, the main advantage of the MADRL controllers lies in their ability to exploit both the electricity price signals and the coupling among different energy resources. The EV charging cost of **Rule-based + Battery** is 386.90 EUR, while its storage profit is only 112.89 EUR. In comparison, the EV charging costs of **Soft Base**, **Soft Base + Progress**, **Base Hard**, and **Projection Hard** are 329.78 EUR, 335.68 EUR, 287.53 EUR, and 373.42 EUR, respectively, all of which are lower than that of **Rule-based + Battery**. At the same time, their storage profits are 272.39 EUR, 146.98 EUR, 205.87 EUR, and 187.39 EUR, respectively, all exceeding the 112.89 EUR achieved by **Rule-based + Battery**. In other words, MADRL does not simply charge or discharge according to predefined rules. Instead, it learns more flexible temporal relationships among the load demand, PV generation, electricity price, battery SoC, and EV SoC. Therefore, although **Projection Hard** has a relatively high average EV charging price, its overall EV charging cost remains lower than that of the rule-based controller, while its battery profitability is substantially higher.

However, from a purely economic perspective, the MADRL controllers are still inferior to the MPC-based controllers. Under the soft constraint scenario, the EV charging costs of ADMM-MPC Soft and Local MPC Soft are 210.69 EUR and 208.32 EUR, respectively, which are significantly lower than the 329.78 EUR of **Soft Base** and the 335.68 EUR of **Soft Base + Progress**. Under the hard constraint scenario, the EV charging costs of ADMM-MPC Hard and Local MPC Hard are 235.74 EUR and 235.68 EUR, respectively, which are also lower than the 287.53 EUR of **Base Hard** and the 373.42 EUR of **Projection Hard**. The reason is that MPC explicitly utilizes forecasts of future electricity prices, load demand, and PV generation within each rolling optimization horizon and directly solves a constrained optimization problem. Consequently, it is easier for MPC to schedule EV charging during low-price periods. In contrast, the MADRL actor has to learn this scheduling strategy indirectly through training samples and reward feedback. Moreover, during execution, its actions are further influenced by the EV feasibility check and the safety projection. Therefore, its economic performance is generally inferior to that of MPC.

Table 4.10 summarizes the EV departure SoC and distribution network security indicators of the selected controllers and the baseline controllers. From the perspective of EV feasibility, **Soft Base + Progress**, **Projection Hard**, **Rule-based + Battery**, and **Rule-based No Battery** all achieve an average departure SoC of 0.9500, a minimum departure SoC of 0.9500, and zero below-target occurrences. In contrast, **Soft Base** achieves an average departure SoC of 0.9460 and a minimum departure SoC of 0.8782, with 2 below-target occurrences, indicating that the EV departure requirement cannot always be satisfied when only the basic reward function is used. **Base Hard** achieves an average departure SoC of 0.9098 and a minimum departure SoC of 0.9000,

remaining close to the target SoC overall, but still exhibiting one numerically below-target occurrence. In comparison, the average departure SoC of the MPC controllers is approximately 0.896–0.900, which is closer to the minimum required SoC than to the upper SoC limit. This is beneficial for reducing the EV charging cost, but under the evaluation criterion adopted in this thesis, it results in a relatively large number of below-target records.

Table 4.10: Comparison of EV Departure SoC and Grid Safety Metrics among Different Controllers

Controller	Mean dep. SoC	Below target	Voltage vio.	Max trafo loading
Soft Base	0.9460	2	2	195.22%
Soft Base + Progress	0.9500	0	1	136.97%
Base hard	0.9098	1	1	156.26%
Projection Hard	0.9500	0	0	108.54%
ADMM-MPC Soft	0.8961	45	0	97.75%
Local MPC Soft	0.8963	45	19	168.48%
ADMM-MPC Hard	0.9000	45	0	118.94%
Local MPC Hard	0.9000	45	25	163.63%
Rule-based + Battery	0.9500	0	3	128.76%
Rule-based No Battery	0.9500	0	0	125.63%

From the perspective of distribution network security, **Projection Hard** is the most robust controller among the four selected MADRL controllers. It achieves zero voltage violations and a maximum transformer loading of 108.54%, which is substantially better than the 2 voltage violations and the maximum transformer loading of 195.22% under **Soft Base**, as well as the 1 voltage violation and the maximum transformer loading of 156.26% under **Base Hard**. Compared with **Rule-based + Battery**, **Projection Hard** reduces the number of voltage violations from 3 to 0, while decreasing the maximum transformer loading from 128.76% to 108.54%. These results indicate that, although the safety projection sacrifices part of the EV economic performance, it can correct the battery power and PV curtailment before action execution, thereby reducing the risks of voltage violations and transformer overloading.

Compared with MPC, the distribution network security performance of MADRL depends on the controller configuration. ADMM-MPC achieves relatively good distribution network security. For example, ADMM-MPC Soft produces zero voltage violations and a maximum transformer loading of only 97.75%, indicating that the centrally coordinated MPC framework has an advantage in satisfying the network constraints. In contrast, the Local MPC controllers exhibit weaker distribution network security. Local MPC Soft

results in 19 voltage violations with a maximum transformer loading of 168.48%, while Local MPC Hard produces 25 voltage violations with a maximum transformer loading of 163.63%. By comparison, **Projection Hard** achieves zero voltage violations, and its maximum transformer loading is only 108.54%. Therefore, although MADRL is economically inferior to MPC, particularly with respect to the overall cost achieved by Local MPC, the incorporation of the safety projection enables MADRL to obtain more stable distribution network security performance than Local MPC.

Overall, the advantages of the MADRL controllers over **Rule-based + Battery** arise from two aspects. First, MADRL learns electricity price-aware EV charging and battery arbitrage strategies, resulting in lower EV charging costs than the rule-based controller while significantly improving the storage profit. Second, after incorporating the progress penalty or the projection mechanism, MADRL achieves EV departure SoC performance and distribution network security that are comparable to, or even better than, those of the fixed rule-based controller. Compared with MPC, the primary limitation of MADRL lies in its economic performance. MPC directly solves a constrained optimization problem over the prediction horizon, allowing the EV charging schedule to approach the global optimum more closely and thereby reducing the EV charging cost. In contrast, MADRL is jointly influenced by training stability, reward function design, and action projection, which may sacrifice part of the charging opportunities during low-price periods in order to satisfy the safety constraints or the EV SoC requirements.

Therefore, MADRL is clearly superior to the rule-based controllers and demonstrates the capability to regulate the distribution network through intelligent scheduling. When the EV departure feasibility and distribution network security are further considered, **Projection Hard** is the most robust controller among the selected MADRL controllers, whereas **Soft Base + Progress** provides a good trade-off between EV feasibility and economic performance under the soft constraint framework.